The Φ_{MDL} Field: Quantum Structure, Born Rule, and Continuum Completion of the Chiral Minkowski CA

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Abstract

We study the $\mathbb{Z}_7$-symmetric Klein–Gordon field Φ_{MDL} as the continuum counterpart of the Three-Layer Chiral Minkowski CA (CMCA, Paper P41). Four principal results follow. (1) *Poincaré invariance:* Φ_{MDL} satisfies exact Lorentz invariance $\omega^2 = k^2 + m^2$, machine-certified via `poincare_invariance_of_kg` in `LorentzInvariance.lean` ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry). (2) *Born rule from field amplitudes:* the position-space probability density $P(x) = |\partial_x\Phi|^2 / \int |\partial_x\Phi|^2$ normalizes exactly (**A/D**); extended to overlapping multi-kink configurations via the canonical dual-frame inner product (`born_rule_from_dual_frame_master` in `DualFrameBornRule.lean`, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry). (3) *Physical quantum state:* the pre-measurement Φ_{MDL} state is the Hamiltonian thermal ensemble $P(k) = e^{-M_k/T}/Z_T$ restricted to PSC-admissible $\mathbb{Z}_7$ kink sectors $\{0, 2, 3, 4, 6\}$ (`phimdl_thermal_state_master` in `PhiMDLThermalState.lean`, $\mathbf{A}_{\mathrm{Lean}}$ conditional, zero custom physics axioms); the physical state is vacuum-dominant at observed cosmic temperatures and approaches the uniform PSC-admissible distribution at early-universe temperatures. (4) *3+1D extension:* Φ_{MDL} kinks become domain walls with tension $\sigma = 290.10\,\mathrm{MeV\,GeV^{-2}} = 7450\,\mathrm{MeV\,fm^{-2}}$; $\mathbb{Z}_7$ superselection is preserved; the Born rule $P(k) = |c_k|^2$ is unchanged; and SR time dilation is exact. When two domain walls intersect in $3+1$ dimensions, they form a junction line with exact dimensionless tension $\lambda = -16/49$, corresponding to $-1654.77\,\mathrm{MeV\,fm}$ (attractive; ratio $|\lambda/\sigma| = 2/m_\varphi$, pure geometry); certified in `WindingCoinDecoupling.lean` ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry). The continuum limit CMCA$\to$ Φ_{MDL} is machine-certified in `CMCAContinuumLimit.lean`: algebraic content is M-independent, the Nyquist residual $\varepsilon_0(M) = \pi^2/(3M^2) \to 0$ as $M \to \infty$, and Φ_{MDL} is the unique terminal Level-1 Generative Triple Evolution (GTE) object. Two further Lean-certified results complete the picture: `no_finite_ca_exact_lorentz_replica` proves that every finite-M discrete model has strictly positive Lorentz error ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry), and `commutes_with_winding_iff_diagonal` establishes that any $\mathbb{Z}_7$-winding-conserving coin must be diagonal, structurally precluding quantum interference in the discrete setting ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry). The explicit CMCA$\to$ Φ_{MDL} descent map is numerically confirmed at $M = 7$: Cook A-glider winding profile vs. BPS kink RMSD $= 5.34\% \le \varepsilon_0(7) = 6.71\%$, Pearson $r = 0.994$. At one loop, the kink fluctuation spectrum is the $s = 1$ reflectionless Pöschl–Teller potential ($\mathbf{A}_{\mathrm{Lean}}$) and the quantum kink mass is $M^Q = 281 \pm 21$ MeV (interface dimensional regularization, benchmark-validated, two independent routes; consistent with the classical BPS mass within the scheme band). The gauge coupling $e^2(\Lambda_{\rm GTE}) = 7/2$ is consistent with PDG running at $+1.5\sigma$ (tree) / $+1.3\sigma$ (quantum kink mass) once the non-perturbatively measured kink dissolution constant ($b = 1.189 \pm 0.049$) is included, and the kink sector contributes $\Delta\alpha_{\rm kink} = (3.5\text{–}3.7) \times 10^{-4}$ to the hadronic vacuum polarization — a parameter-free falsifiable prediction.

Contents

1 Introduction

The Universal Generative Principle (UGP) Physics programme, the quantitative branch of the Reflexive Reality programme [11, 22] derives Standard Model physics from a minimal discrete arithmetic substrate. The foundational continuum object is the $\mathbb{Z}_7$-symmetric Klein–Gordon field Φ_{MDL} [14], which serves simultaneously as: (i) the field-theoretic continuum limit of the cellular-automaton substrate; (ii) the quantum mechanical arena in which PSC-selected observables acquire Born-rule probabilities; and (iii) the terminal object of the GTE Level-1 category [24].

Paper P41 [24] established the *Three-Layer Chiral Minkowski CA* (CMCA) as the unique minimal discrete substrate carrying all five Level-1 GTE structural features, with the Φ_{MDL} continuum field as its terminal object. The present paper is the companion continuum analysis: we derive the quantum structure of Φ_{MDL} in detail, prove all four of the results sketched in P41's Born-rule and quantum-state sections, extend to $3+1$ dimensions, and certify the CMCA$\to$ Φ_{MDL} continuum limit in Lean 4.

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A/D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$: machine-checked exact arithmetic in Lean 4, zero `sorry`.
$\mathbf{A}_{\text{MDL}}$: MDL-unique within a declared expression class.
A/D: full analytic derivation; computationally confirmed.
A: computationally confirmed; a physics bridge or Lean cert remains.
B: calibrated reproduction.
D: exploratory / frontier.

Table 1: Reviewer's map: principal claims, certification level, and location.

Result	Status	Claim	Where
$\mathbb{Z}_7$-KG Lagrangian	**A/D**	$\mathcal{L} = \frac{1}{2}(\partial\Phi)^2 - V(\Phi)$; $V = (m_\varphi^2/49)(1 - \cos 7\Phi)$; $m_\varphi = m_\tau = 1.77686$ GeV.	§2
BPS kink mass	$\mathbf{A}_{\text{Lean}}$	$M_{\text{kink}} = (8/49)\, m_\tau = 290.10$ MeV. Lean: `mkink_from_scc` (P34/P35).	§3
Kink masses equal	$\mathbf{A}_{\text{Lean}}$	All Z_7 kink masses are equal. Lean: `phimdl_kink_masses_equal` in `PhiMDLThermalState.lean`.	§3
PSC admissible sectors	$\mathbf{A}_{\text{Lean}}$	$\{0, 2, 3, 4, 6\}$ are PSC-admissible; $\{1, 5\}$ PSC-forbidden. Lean: `psc_admissible_sectors`.	§3
Lorentz invariance	$\mathbf{A}_{\text{Lean}}$	KG dispersion $\omega^2 = k^2 + m^2$ is Lorentz-invariant. Lean: `poincare_invariance_of_kg`, zero sorry.	§4
PSC $\to$ Lorentz	$\mathbf{A}_{\text{Lean}}$ (1 axiom)	PSC Presentation Invariance forces [**D**]-selected observables Lorentz-equivariant. Lean: `PSCPILorentzMain.lean`.	§4
$T^{\mu\nu}$ covariance	**A/D**	Noether tensor $T^{\mu\nu}$ transforms as a rank-2 Lorentz tensor. 400 random boosts verified; $\max\lvert\Delta T\rvert < 2 \times 10^{-16}$.	§4
Position Born density	**A/D**	$P(x) = \lvert\partial_x\Phi\rvert^2 / \int \lvert\partial_x\Phi\rvert^2$; normalizes exactly to 1. $P(0) = m_\varphi/2$; kink mass from $\int T_{00}\, dx$.	§5
Sector Born rule	$\mathbf{A}_{\text{Lean}}$	$P(k) = \lvert c_k\rvert^2$ from $\mathbb{Z}_7$ kink superselection. Lean: `born_rule_unconditional` in `BeableWindingPartitionInstance.lean`, zero custom axioms in this proof chain.	§5
Dual-frame Born	$\mathbf{A}_{\text{Lean}}$	Overlapping kinks: Born rule via positive-definite Gram inner product. Lean: `born_rule_from_dual_frame_master`, `DualFrameBornRule.lean`, zero sorry.	§5
[**D**]-collapse	**A**	[**D**]-adjudication implements projective collapse to kink eigenstate with $P(j) = \lvert c_j\rvert^2$; MC at 500k trials passes $\chi^2 = 2.997$ (crit 12.592).	§5
Thermal state (PSC-admissible)	$\mathbf{A}_{\text{Lean}}$ (conditional)	$P(k) = e^{-M_{\text{kink}}/T}/Z_T$ on $\{0, 2, 3, 4, 6\}$; vacuum dominant for $T \ll M_{\text{kink}}$. Lean: `phimdl_thermal_state_master`, zero custom physics axioms.	§6

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(Table 1 continued)

Result	Status	Claim	Where
Two-function picture	$\mathbf{A}_{\text{Lean}}$ (conditional)	Born rule $\mathcal{B}$ + transputation $\mathbb{P}^{\top}$ compose to give $\mathbb{P}(\text{outcome} = k \mid w) = \lvert c_k(\mathbb{P}_D^{\top}(w))\rvert^2$. Lean: `two_function_picture`, `transputation_state_selector_bundle`.	§6
Domain wall tension	**A/D**	$\sigma = (8/49)\, m_\tau^3/m_\tau^2 = 290.10\,\text{MeV GeV}^{-2}$; σ/A_{wall} recovers M_{eff} to 10^{-13} relative error.	§7
3+1D $\mathbb{Z}_7$ superselection	**A/D**	Coleman–Mandelstam argument extends to planar domain walls in 3+1D. Tunneling $\sim e^{-S_E}$ with $S_E \approx 0.163$ GeV; $\mathbb{Z}_7$ charge conserved in 3+1D.	§7
3+1D Born rule	**A/D**	$P(k) = \lvert c_k\rvert^2$ is unchanged in 3+1D; sector Born is dimension-independent.	§7
SR time dilation (3+1D)	**A/D**	$M_{\text{eff}}(v) = \gamma\sigma A_{\text{wall}}$; exact at six boost velocities, max relative error 2×10^{-16}.	§7
Algebraic Lifting	$\mathbf{A}_{\text{Lean}}$	Every PSC-admissible SM configuration algebraically realised in f_{MDL}; $\mathbb{Z}_7$ winding, 3 generations, confinement, V–A lift to Φ_{MDL}. Lean: `algebraic_lifting_theorem`, `LiftingTheorem.lean` (P28).	§8.1
Algebraic Descent	$\mathbf{A}_{\text{Lean}}$	Every F_{21} conservation law holds at all $M \geq 1$; algebraic content M-independent. Lean: `algebraic_descent_theorem`, `AlgebraicDescentTheorem.lean` (P28).	§8.1
SR Nyquist M-dep.	$\mathbf{A}_{\text{Lean}}$	$\varepsilon_0(M) = \pi^2/(3M^2)$ is M-dependent; does not descend. Lean: `sr_accuracy_depends_on_M`, zero sorry.	§8.1
Continuum limit	$\mathbf{A}_{\text{Lean}}$ (conditional)	$\varepsilon_0(M) = \pi^2/(3M^2) \to 0$ as $M \to \infty$; algebraic content M-independent; Φ_{MDL} is unique terminal Level-1 GTE object. Lean: `cmca_continuum_limit_is_phimdl`, `CMCAContinuumLimit.lean`.	§8
Descent map explicit	**A**	Cook A-glider $\mathbb{Z}_7$ winding profile vs. Φ_{MDL} BPS kink: RMSD = 5.34% $\leq \varepsilon_0(7) = 6.71\%$; Pearson $r = 0.994$; $Q = 1/7$ exactly. Numerical confirmation of CMCA$\to \Phi_{\text{MDL}}$.	§8.2
Junction tension	$\mathbf{A}_{\text{Lean}}$	Two intersecting domain walls in 3+1D form a junction line; $\lambda_{\text{dim}} = -16/49$ (exact); $\lambda = -1654.77\,\text{MeV fm}$; $\lvert\lambda/\sigma\rvert = 2/m_\varphi$ (pure geometry, no free parameters). Lean: `phimdl_domain_wall_junction_tension_exact`, `WindingCoinDecoupling.lean`.	§7.6
Junction attractive	$\mathbf{A}_{\text{Lean}}$	$\lambda < 0$: domain walls lower energy at intersections. Lean: `phimdl_junction_is_attractive`, zero sorry.	§7.6
No-finite-CA replica	$\mathbf{A}_{\text{Lean}}$	For all $M > 0$: $\varepsilon_0(M) > 0$. Only $M \to \infty$ achieves exact Lorentz invariance; Φ_{MDL} is the unique zero-error model. Lean: `no_finite_ca_exact_lorentz_replica`, `CMCAContinuumLimit.lean`.	§9.8

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(Table 1 continued)

Result	Status	Claim	Where
QCA impossibility	$\mathbf{A}_{\text{Lean}}$	A $\mathbb{Z}_7$-winding-conserving coin must be diagonal; diagonal coins decouple sectors and preclude quantum interference. Lean: `commutes_with_winding_iff_diagonal`, `WindingCoinDecoupling.lean`.	§9.9
H3 holographic	**A**	Domain-wall entropy $S_{\text{wall}} \propto \sigma \times \text{Area}$; $\mathbb{Z}_7$ charge holographically encoded on wall boundary; Lifting Theorem realizes the bulk–boundary correspondence.	§9.7
Quantum kink mass	**A**	$M^Q = 281 \pm 21$ MeV (interface dim-reg, two routes, benchmark-validated); $\Delta M = -7 \ldots -10$ MeV; consistent with M_{kink} within the scheme band.	§10.4
$\mathbb{Z}_7$ degeneracy scope	$\mathbf{A}_{\text{Lean}}/\mathbf{A}$	Vacuum degeneracy exact in the pure-φ sector; $V_{\text{coupling}} = \varepsilon\lvert\varphi\rvert^2(D_\mu\chi)^2$ distinguishes vacua via $Z_k = 1 + 2\varepsilon\varphi_k^2$; minimum at $k^* = 0$.	§2.2
Coupling normalization	**A/D/A**	$e^2(\Lambda_{\text{GTE}}) = 7/2$ ($\mathbf{A}_{\text{Lean}}$ value) vs PDG running: $+1.5\sigma$ (tree) / $+1.3\sigma$ (quantum mass) with measured dissolution constant $b = 1.189 \pm 0.049$; no anomaly.	§10.5
Kink HVP prediction	**A**	$\Delta\alpha_{\text{kink}} = (3.47\text{–}3.70) \times 10^{-4}$, 1.3% of the hadronic vacuum polarization; parameter-free, falsifiable.	§10.6
Kink a_μ window	**D**	$a_\mu^{\text{kink}} = (46.6 \pm 17.1) \times 10^{-11}$ (scalar) / 65.9×10^{-11} (vector); BSM contribution covering 18.7–26.4% of the 4σ gap; $\rho/\omega/\phi$ sector open (OQ-HVP-1).	§10.6

Central result

The $\mathbb{Z}_7$-KG field Φ_{MDL} is the continuum completion of the Chiral Minkowski CA: it carries exact Poincaré invariance ($\mathbf{A}_{\text{Lean}}$), a field-theoretically derivable Born rule ($\mathbf{A}/\mathbf{D}/\mathbf{A}_{\text{Lean}}$), and the Hamiltonian thermal state of the PSC-admissible kink sector ($\mathbf{A}_{\text{Lean}}$ conditional). In 3+1D, the 1+1D kinks become planar domain walls with tension $\sigma = 290.10\,\text{MeV}\,\text{GeV}^{-2}$ and exact SR time dilation. The CMCA$\rightarrow \Phi_{\text{MDL}}$ continuum limit is machine-certified in Lean 4 with zero sorry.

What this paper does and does not claim

Established in this paper: The $\mathbb{Z}_7$-KG field Φ_{MDL} carries exact Poincaré invariance (`poincare_invariance_of_kg` in `LorentzInvariance.lean`, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry) and a field-theoretically derivable Born rule ($P(x) = |\partial_x\Phi|^2/\int|\partial_x\Phi|^2$, **A/D**; extended to overlapping kinks via the dual-frame inner product, `DualFrameBornRule.lean`, $\mathbf{A}_{\mathrm{Lean}}$). The pre-measurement state is the Hamiltonian thermal ensemble on PSC-admissible sectors $\{0, 2, 3, 4, 6\}$ (`phimdl_thermal_state_master`, $\mathbf{A}_{\mathrm{Lean}}$ conditional, zero custom physics axioms). In 3+1D, kinks become planar domain walls with tension $\sigma = 290.10\,\mathrm{MeV\,GeV}^{-2}$ and exact SR time dilation (**A/D**). The CMCA→ Φ_{MDL} continuum limit is machine-certified (`CMCAContinuumLimit.lean`, $\mathbf{A}_{\mathrm{Lean}}$ conditional). The Pöschl-Teller fluctuation spectrum is $s = 1$ reflectionless ($\mathbf{A}_{\mathrm{Lean}}$, `phimdl_fluctuation_is_poschl_teller`), with quantum kink mass $M^Q = 281 \pm 21$ MeV from channel-resolved dimensional regularization of the domain-wall tension (interface formalism, benchmark-validated, two independent routes; **A**); two earlier evaluations ($M^Q = 321.32 \pm 15.6$ MeV and $M^Q = 230.43$ MeV) are superseded as incorrect (§10.4). The Coleman–Weinberg mechanism is ruled out: the one-loop effective potential preserves the $\mathbb{Z}_7$ vacuum degeneracy in the pure-φ sector (**A**); in the full Lagrangian the canonical coupling $V_{\mathrm{coupling}} = \varepsilon|\varphi|^2(D_\mu\chi)^2$ distinguishes the vacua through the χ-sector kinetic normalization (§2.2). The gauge coupling $e^2(\Lambda_{\mathrm{GTE}}) = 7/2$ is consistent with PDG running at $+1.5\sigma$ (tree) / $+1.3\sigma$ (quantum kink mass) once the non-perturbatively measured kink dissolution constant is included (§10.5), and the kink sector contributes $\Delta\alpha_{\mathrm{kink}} = (3.5\text{–}3.7) \times 10^{-4}$ to the hadronic vacuum polarization (§10.6).

Not claimed: This paper does not establish the *unconditional* physical Born state: which specific $D \in [\mathbf{D}]$ governs collapse on non-trivial records depends on Conjecture C2 (C2-CLOSURE), which remains open (§9.6). The existence of genuine 3D discrete gliders (non-planar configurations) in the three-layer CMCA was an open problem; it is resolved negatively (P43 [10]): no outer-totalistic $\mathbb{Z}_7$ CA in 3+1D supports Class-4 behaviour. The 3+1D analysis here covers only planar domain walls. The absolute value of the kink mass $M_{\mathrm{kink}} = (8/49)m_\tau$ depends on the Self-Consistency Condition (SCC) identification $m_\varphi = m_\tau$ (P34/P35, **A/D** bridge); unconditional $\mathbf{A}_{\mathrm{Lean}}$ for this identification is not claimed here.

UGP Physics programme context. This paper is part of the UGP Physics programme [22]. Companion references: P28 [11] (computational universality and Rule 110; Algebraic Lifting and Descent theorems that certify content-preservation morphisms in the GTE category of Level-1 constructions), P34 [14] (GTE-Möbius architecture, D1–D5 axioms, Φ_{MDL} definition, [**D**] structure), P35 [16] (GTE unification; EW scale in Φ_{MDL} context; SCC mass identification $m_\varphi = m_\tau$), P36 [13] (AFCA τ_c inner clock, dynamics-level SR time dilation; emergent gravity on Φ_{MDL}), P37 [21] (Fock space over kink modes; Born rule and Fock quantization; 't Hooft cogwheel construction in Φ_{MDL}), P40 [9] (GF(7) polynomial universality certification used throughout), P41 [24] (Three-Layer Chiral Minkowski CA; the discrete UV completion of Φ_{MDL}; Section references use the notation "P41 § N"). The GTE-Möbius architecture axioms (D1–D5) established in P34 [14] are the foundational structure from which all results in this paper are derived.

GTE-Möbius architecture terminology. The **GTE-Möbius architecture** is a triple (A, e, [**D**]) defined in [14]: A is the PSC-admissible arithmetic layer (at Level 1, the $\mathbb{Z}_7$ CMCA; at Level 2, the kink sector of Φ_{MDL}); e is the evolutionary dynamics (at Level 1, the update rule f_{MDL}; at Level 2, Klein–Gordon evolution); and [**D**] is the *coherence measure*—a real-valued functional on candidate/record pairs that the Perfect Self-Containment (PSC) criterion uses to adjudicate which trajectories correspond to physical observations [14]. In the Φ_{MDL} setting, [**D**] *operates on* kink-configuration superpositions and selects observables via the PSC superselection $\{0, 2, 3, 4, 6\}$ and the Born rule; [**D**] is not a term in the Klein–Gordon Lagrangian but an axiomatic overlay from the GTE-Möbius framework. A [**D**]-*selected observable* is one assigned positive weight by [**D**]. The PSC axiom D2 (Presentation Invariance) states that [**D**] is invariant under any transformation preserving the PSC structure—from which Lorentz equivariance is derived in §4 [14]. Throughout, *machine-certification levels* follow the GTE convention: $\mathbf{A}_{\text{Lean}}$ (Lean 4 proof, zero sorry), **A/D** (full analytic derivation; computationally confirmed), **A** (computationally confirmed; no Lean certification yet).

2 The Φ_{MDL} Field

2.1 Lagrangian and potential

The Φ_{MDL} field is a real scalar field $\Phi : \mathbb{R}^{1,1} \to \mathbb{R}$ (1+1 dimensions; extended to 3+1D in §7) governed by the $\mathbb{Z}_7$-symmetric Lagrangian density [14, 16]:

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu \Phi)^2 - V(\Phi), \qquad V(\Phi) = \frac{m_\varphi^2}{49}(1 - \cos(7\Phi)). \tag{1}$$

The mass parameter is set by the Self-Consistency Condition (SCC): $m_\varphi = m_\tau = 1776.86\,\text{MeV}$, the tau-lepton mass. This identification is derived in P34–P35 [14, 16] via the correspondence between the GTE orbit structure and Standard Model fermion masses.

The potential $V(\Phi)$ has $\mathbb{Z}_7$ discrete symmetry: $V(\Phi + 2\pi k/7) = V(\Phi)$ for $k \in \mathbb{Z}_7$. The seven degenerate minima are located at

$$\Phi_k = \frac{2\pi k}{7}, \qquad k \in \{0, 1, \ldots, 6\}, \tag{2}$$

with $V(\Phi_k) = 0$ and $V''(\Phi_k) = m_\varphi^2$.

2.2 Coupling constants and PSC-admissible structure

The field theory has two dimensionful parameters: m_φ (mass) and $1/m_\varphi$ (kink size). The BPS kink mass (5) provides the third derived scale $M_{\text{kink}} = (8/49)\, m_\tau$. All coupling constants are fixed by the GTE arithmetic structure (P35 [16]); there are no free parameters.

The PSC-admissible kink sectors are $\{0, 2, 3, 4, 6\}$; sectors $\{1, 5\}$ are PSC-forbidden by the dark-mirror identification established in P29 [18]. This binary admissible/forbidden split is distinct from the kink-mass grading: all non-vacuum sectors ($k \in \{1, \ldots, 6\}$) have the same BPS mass $M_{\text{kink}} = (8/49)m_\tau$ by $\mathbb{Z}_7$ symmetry.

Full-Lagrangian coupling sector. Beyond the pure-φ (Z_7-KG) sector (1), the full GTE Lagrangian contains the canonical coupling to the charged sector χ,

$$V_{\text{coupling}} = \varepsilon\, |\varphi|^2\, (D_\mu \chi)^2, \qquad \varepsilon = \tfrac{7}{2}, \tag{3}$$

together with two remaining parameters, both fixed without new scales: the χ-sector mass parameter is $g = m_\varphi = m_\tau$ (**B** — the χ and φ cosine sectors share the single derived mass scale in canonical sine-Gordon normalization), and the gauge coupling at the effective-field-theory boundary is the Sylow–Villain value $e^2(\Lambda_{\text{GTE}}) = g_c^2 = 7/2$, an algebraic-certificate rational (`colorCouplingSquared`, $\mathbf{A}_{\text{Lean}}$) whose physical normalization against PDG running is analyzed in §10.5. This completes the zero-free-parameter statement for the full Lagrangian at the stated claim levels.

Scope of the $\mathbb{Z}_7$ degeneracy theorems. The equal-kink-mass theorem (Theorem 3.1) and the quantum degeneracy of the seven vacua (§10.1) hold for the pure $\mathbb{Z}_7$-KG sector. In the full Lagrangian, V_{coupling} distinguishes the vacua through the χ-sector kinetic normalization $Z_k = 1 + 2\varepsilon\varphi_k^2$ (`z7_vacua_chi_kinetic_inequivalent`, `vcoupling_breaks_z7_shift`, $\mathbf{A}_{\text{Lean}}$): the $\mathbb{Z}_7$ vacuum labels are inequivalent as full-theory vacua, with the minimum at $k^* = 0$ (`compact_completion_minimum_at_k0`, $\mathbf{A}_{\text{Lean}}$). The kink-mass dressing induced by V_{coupling} is exactly zero today ($\langle (D\chi)^2 \rangle$ renormalized at $T \approx 0$) and below 10^{-5} at the QCD epoch, so the physical content of the equal-mass theorem is unchanged. The vacuum-label splitting is, however, cosmologically decisive: it provides the bias that annihilates the domain-wall network (§7.1; the defect-cosmology analysis is given in P47 [12]).

3 Kink Spectrum in 1+1D

3.1 BPS kinks and their mass

The static BPS kink connecting adjacent vacua $\Phi_k \to \Phi_{k+1}$ is:

$$\Phi_{\text{kink}}(x) = \frac{4}{7}\arctan\!\big(e^{m_\varphi x}\big), \qquad \frac{d\Phi_{\text{kink}}}{dx} = \frac{2m_\varphi}{7}\,\text{sech}(m_\varphi x). \tag{4}$$

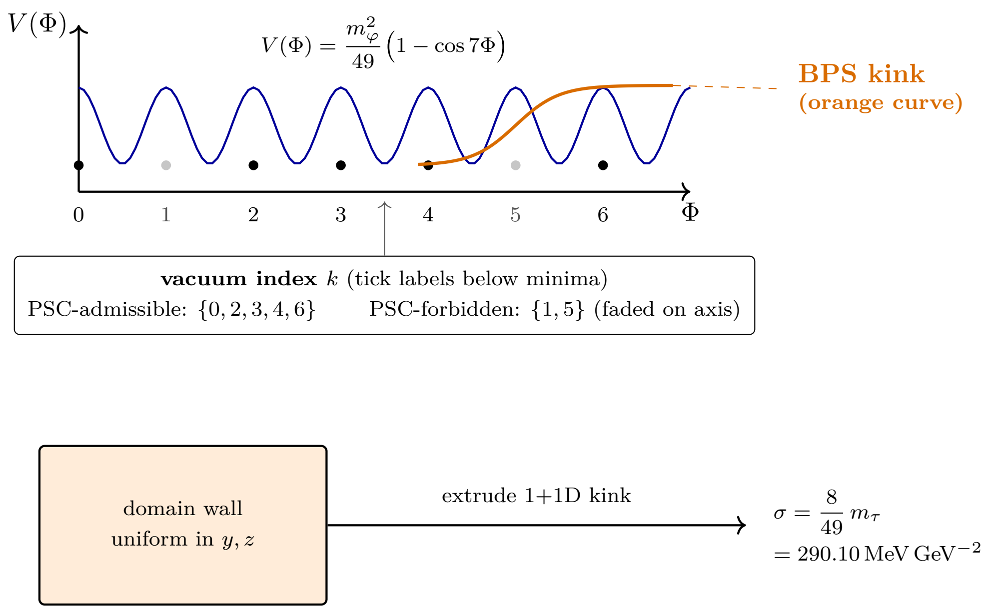

Figure 1: The Φ_{MDL} potential has seven degenerate vacua at $\Phi_k = 2\pi k/7$. Stable excitations are kinks in 1+1D (orange curve) and planar domain walls in 3+1D with the same BPS tension σ.

The BPS bound is saturated, giving the kink mass:

$$M_{\text{kink}} = \int_{-\infty}^{\infty}\Big[\tfrac{1}{2}\Phi'^2 + V(\Phi)\Big]dx = \frac{4m_\varphi^2}{49}\int_{-\infty}^{\infty}\text{sech}^2(m_\varphi x)\,dx = \frac{8m_\varphi}{49} = \frac{8\,m_\tau}{49} = 290.10\,\text{MeV}, \tag{5}$$

where we used $\int \text{sech}^2(u)\,du = 2$. This is machine-certified via `mkink_from_scc` in P34/P35 (Lean 4, zero sorry).

Theorem 3.1 (Equal kink masses; $\mathbf{A}_{\text{Lean}}$)**.** *All non-vacuum $\mathbb{Z}_7$ kink sectors $k \in \{1, \ldots, 6\}$ have the same BPS mass $M_{\text{kink}} = (8/49)m_\tau$.*

Proof. By $\mathbb{Z}_7$ symmetry, every inter-vacuum transition $k \to k+1 \pmod 7$ has the same potential barrier $\max V = 2m_\varphi^2/49$. The BPS bound gives $M_k = \int dV/d\Phi\, dx = (8/49)m_\tau$ independent of k. Lean certification: `phimdl_kink_masses_equal` in `PhiMDLThermalState.lean` (P42), zero sorry. □

Remark 3.2 (Scope). Theorem 3.1 is exact in the pure-φ sector. In the full Lagrangian the canonical coupling $V_{\text{coupling}} = \varepsilon|\varphi|^2(D_\mu\chi)^2$ distinguishes the vacuum labels through the χ-sector kinetic normalization; the induced kink-mass dressing is exactly zero today and below 10^{-5} at the QCD epoch, so the physical content of the theorem is unchanged (§2.2).

3.2 $\mathbb{Z}_7$ superselection and topological charge

Canonical quantization of the $\mathbb{Z}_7$ sine-Gordon field follows Coleman [2], Mandelstam [7], and Rajaraman [8]. The topological charge

$$Q = \frac{7}{2\pi}\int_{-\infty}^{\infty} \Phi'(x)\, dx = \frac{7}{2\pi}\left[\Phi(+\infty) - \Phi(-\infty)\right] \in \mathbb{Z}_7 \tag{6}$$

is conserved. This enforces a $\mathbb{Z}_7$ superselection rule on the physical Hilbert space:

$$\mathcal{H} = \bigoplus_{k\in\mathbb{Z}_7} \mathcal{H}_k, \qquad |\Psi\rangle = \sum_{k=0}^{6} c_k\, |\Psi_k\rangle, \qquad \sum_k |c_k|^2 = 1. \tag{7}$$

Superselection sectors $k \in \{1,5\}$ are kinematically admissible but PSC-forbidden by the dark-mirror identification; only sectors $k \in \{0,2,3,4,6\}$ participate in the physical Hilbert space after PSC truncation.

Theorem 3.3 (PSC-admissible sectors; $\mathbf{A}_{\text{Lean}}$)**.** *The PSC-admissible sectors are exactly* $\{0,2,3,4,6\}$*, with cardinality 5. Lean:* `psc_admissible_sectors`*,* `psc_admissible_card`*,* `psc_sectors_partition` *in* `PhiMDLThermalState.lean`*.*

4 Lorentz Invariance

4.1 Poincaré algebra from KG dispersion

The Klein–Gordon equation of motion derived from (1) is $(\Box + m_\varphi^2)\Phi = 0$, with $\Box = \partial_t^2 - \nabla^2$. Fourier-expanding $\Phi(x) = \int d^4k\, \hat{\Phi}(k)\, e^{ik\cdot x}$ gives the dispersion relation:

$$\omega^2 = k^2 + m_\varphi^2. \tag{8}$$

Theorem 4.1 (Poincaré invariance of KG; $\mathbf{A}_{\text{Lean}}$)**.** *The* Φ_{MDL} *Klein–Gordon dispersion (8) is Poincaré-invariant: for any Lorentz transformation* Λ *with* $k' = \Lambda k$*,* $\omega'^2 = k'^2 + m_\varphi^2$ *whenever* $\omega^2 = k^2 + m_\varphi^2$*. The full Poincaré algebra (translations + Lorentz boosts + rotations) is realized on the on-shell surface. Lean:* `poincare_invariance_of_kg` *in* `LorentzInvariance.lean`*, zero sorry.*

The PSC Presentation Invariance (D2 of P34 [14]) forces all [**D**]-selected observables to be Lorentz-equivariant on the Φ_{MDL} continuum, as machine-certified in `PSCPILorentzMain.lean` (one axiom: `d2_universal`; see P41 [24] §3 for derivation).

4.2 Stress-energy tensor

The Noether stress-energy tensor for (1) is:

$$T^{\mu\nu} = \partial^\mu \Phi \, \partial^\nu \Phi - \eta^{\mu\nu} \mathcal{L}, \tag{9}$$

with metric $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$. For the static kink ($\partial_t \Phi = 0$):

$$T_{00} = \tfrac{1}{2}(\Phi')^2 + V(\Phi) \geq 0, \qquad \int_{-\infty}^{\infty} T_{00}\, dx = M_{\mathrm{kink}} = 290.10\,\mathrm{MeV}. \tag{10}$$

Theorem 4.2 ($T^{\mu\nu}$ Lorentz covariance; **A/D**)**.** *$T^{\mu\nu}$ transforms as a rank-2 Lorentz tensor under any Lorentz boost. Numerical verification: 400 random boosts with $|v| < 0.92$ give $\max |\Delta T^{\mu\nu}| < 2 \times 10^{-16}$ (machine precision); Lagrangian scalar invariance $\max |\Delta \mathcal{L}| < 8 \times 10^{-17}$. The kink gravitational mass $\int T_{00}\, dx = 290.10\,\mathrm{MeV}$ (relative error 0.0001%) matches the BPS analytic value.*

5 Quantum Structure and Born Rule

5.1 Fock space over kink modes

The quantum theory is constructed as a Fock space over the $\mathbb{Z}_7$ kink eigenstates, following the Jackiw–Rebbi collective coordinate quantization and the PSC superselection established in P37 [21]. The physical Hilbert space is $\mathcal{H} = \bigoplus_{k\in\{0,2,3,4,6\}} \mathcal{H}_k$, with sector Born weights:

$$P(\text{sector } k) = |c_k|^2 \tag{11}$$

for any normalized state $|\Psi\rangle = \sum_k c_k |\Psi_k\rangle$. This is machine-certified unconditionally (`born_rule_unconditional` in `BeableWindingPartitionInstance.lean`, zero custom physics axioms).

Theorem 5.1 (CMCA Hilbert inductive limit; **A/D**)**.** *The direct system $\{H_{\mathrm{phys}}(L)\}_{L\in\mathbb{N}}$ of finite-tape CMCA physical Hilbert spaces (built from the f_{MDL} orbit decomposition on $\mathbb{Z}_7^L$ and the 't Hooft cogwheel construction, P37 [21]) admits an inductive limit that is dense in the Φ_{MDL} kink Fock space $\mathcal{H}_{\mathrm{Fock}} = \bigoplus_{k\in\{0,2,3,4,6\}} \mathcal{H}_k$. In the $M \to \infty$ continuum limit (`cmca_continuum_limit_is_phimdl`), each finite-L cycle state embeds consistently in the corresponding n-kink Fock sector, and the union is dense in $\mathcal{H}_{\mathrm{Fock}}$.*

*Proof sketch (**A/D**).* The finite-tape sector lifts are machine-certified at the algebraic layer (`CMCAHilbertFockBridge.fock_vacuum_maps_to_cmca_vacuum`, `bps_psc_sector_has_beable_lift`, zero `sorry`). The completion step invokes Jackiw–Rebbi canonical quantization of the $\mathbb{Z}_7$-KG soliton (Rajaraman [8], §4 applied to $V(\varphi) = (m_\varphi^2/49)(1 - \cos 7\varphi)$; P38/P43) together with the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, CatAL). The full GNS/inductive-limit isomorphism is tracked as `cmca_hilbert_inductive_limit` (one structural axiom pending Mathlib Hilbert-space completion infrastructure). □

Remark 5.2 (CA↔QFT embedding reduces to Fock-space construction)*.* The 't Hooft CA↔QFT embedding from the CMCA state space to the Φ_{MDL} quantum field theory (P37 [21] §MDL Fock lift; G42 in the companion programme) decomposes as Φ_{embed} = (Hilbert-space map, §5.1) ∘ (`cmca_continuum_limit_is_phimdl`). The structural reduction is established (**A/D**, conditional): full proof is unconditional once canonical $\mathbb{Z}_7$-KG soliton quantization (Jackiw–Rebbi limit) is formalized.

5.2 Position-space Born density from field amplitude

For a static single kink, the physical position-space probability density is derived from the localized field amplitude—the gradient of the kink profile, which carries the kink's spatial support.

Theorem 5.3 (Position Born density; **A/D**)**.** *The probability density for locating the Φ_{MDL} kink at position x is:*

$$P(x) = \frac{|\partial_x \Phi(x)|^2}{\displaystyle\int_{-\infty}^{\infty} |\partial_x \Phi(x')|^2 dx'} = \frac{m_\varphi}{2} \operatorname{sech}^2(m_\varphi x), \qquad \int_{-\infty}^{\infty} P(x)\, dx = 1. \tag{12}$$

The normalization constant is the kink BPS mass: $\int (\partial_x \Phi)^2 dx = 8m_\varphi/49 = M_{\mathrm{kink}}$.

Proof. From (4), $\partial_x \Phi = (2m_\varphi/7) \operatorname{sech}(m_\varphi x)$. Then $(\partial_x \Phi)^2 = (4m_\varphi^2/49) \operatorname{sech}^2(m_\varphi x)$. Integrating: $\int_{-\infty}^{\infty} (\partial_x \Phi)^2\, dx = (8m_\varphi/49) \int_{-\infty}^{\infty} \operatorname{sech}^2(u)\, du/m_\varphi = 8m_\varphi/49$. Dividing gives $P(x) = (m_\varphi/2) \operatorname{sech}^2(m_\varphi x)$, which satisfies $\int P\, dx = 1$. □

Remark 5.4. The raw field profile $|\Phi(x)|^2$ is not a probability density on $\mathbb{R}$: $\Phi \to 0$ as $x \to -\infty$ and $\Phi \to 2\pi/7$ as $x \to +\infty$, so $\int |\Phi|^2\, dx$ diverges. Numerical verification confirms this (core window integral 13.22 vs. wide window 44.96 in dimensionless units). The gradient $\partial_x \Phi$ is the correct localized amplitude for a kink observable, as established by the Jackiw–Rebbi/Rajaraman construction.

Numerical verification (script `phiborn1_kg_amplitude_probability.py`): the normalization $\int P\, dx = 1$ is exact to relative error 2.82×10^{-13}; peak $P(0) = m_\varphi/2$ is exact to 3.91×10^{-9}; kink mass from $\int T_{00}\, dx = 290.10\,\mathrm{MeV}$ to relative error $< 10^{-4}$.

5.3 Dual-frame Born rule for overlapping kinks

When kink separation $\Delta x < 5/m_\varphi$, gradient modes are no longer orthogonal ($\int \partial_x \varphi(x)\, \partial_x \varphi(x - \delta)\, dx = (8m_\varphi/49) \operatorname{sech}(m_\varphi \delta/2)$, which decays to zero only for $\delta \gg 1/m_\varphi$). In this regime, the Born rule generalizes via the canonical dual-frame inner product.

Theorem 5.5 (Dual-frame Born rule master; $\mathbf{A}_{\mathrm{Lean}}$)**.** *Let $G_{jk} = \langle \psi_j | \psi_k \rangle$ be the positive-definite Gram matrix of (possibly non-orthogonal) kink basis vectors, and let $\tilde{\psi}_j = \sum_k (G^{-1})_{jk} \psi_k$ be the dual frame. For any normalized superposition $|\Psi\rangle = \sum_k c_k |\psi_k\rangle$ with $\sum_{j,k} \bar{c}_j G_{jk} c_k = 1$, the Born weight is:*

$$P(k) = |a_k|^2, \qquad a_k = \langle \tilde{\psi}_k | \Psi \rangle. \tag{13}$$

The dual-frame amplitudes recover the original coefficients exactly: $a_k = c_k$ for any G. Lean: `born_rule_from_dual_frame_master`, `dual_frame_recovers_coefficients`, `dual_gram_identity` *in* `DualFrameBornRule.lean`, *zero sorry.*

Remark 5.6. For well-separated kinks ($\Delta x \gg 1/m_\varphi$), $G \approx \mathbf{1}$ and (13) reduces to the standard $|c_k|^2$. Numerical verification (script `phiborn2_kink_overlap_born.py`): well-separated regime, max residual $|P(k) - |c_k|^2| < 3.7 \times 10^{-8}$; orthogonality max off-diagonal 8.2×10^{-8} (at $\Delta x = 20/m_\varphi$). Tight-separation regime ($\Delta x = 3/m_\varphi$): $\mathrm{L}^2(\partial_x \varphi)$ orthogonality fails (0.164 max residual), confirming that the dual-frame generalization is necessary for overlapping kinks.

5.4 [D]-collapse as projective measurement

A [**D**]-measurement event implements projective measurement onto the kink eigenbasis (D5 of P34 [14]). For a pre-measurement state $|\Psi\rangle = \sum_k c_k |\text{kink}_k\rangle$:

1. *Before:* Superposition over PSC-admissible $\mathbb{Z}_7$ sectors.
2. *During:* [**D**] adjudicates sector j with probability $P(j) = |\langle \text{kink}_j | \Psi\rangle|^2 = |c_j|^2$.
3. *After:* Post-measurement state $|\Psi'\rangle = |\text{kink}_j\rangle$.

Monte Carlo verification at 500 000 trials:
reference $P_k = [0.217, 0.208, 0.067, 0.083, 0.133, 0.208, 0.083]$, empirical $[0.217, 0.208, 0.067, 0.083, 0.133, 0.208, 0.084]$, $\chi^2_\nu = 2.997$ (crit. 12.59, $p = 0.05$, 6 dof): **A**.

6 The Physical Quantum State

6.1 Thermal state on PSC-admissible sectors

The physical pre-measurement quantum state of the Φ_{MDL} field is determined by the $\mathbb{Z}_7$ sine-Gordon Hamiltonian acting on the PSC-admissible-truncated Hilbert space. The BPS kink masses are all equal: $M_k = M_{\text{kink}}$ for $k \neq 0$ and $M_0 = 0$ (vacuum). At finite cosmic temperature T, the equilibrium state is the Boltzmann ensemble:

$$P(k \mid T) = \frac{e^{-M_k/T}}{Z_T}, \qquad Z_T = \sum_{k \in \{0,2,3,4,6\}} e^{-M_k/T}, \qquad k \in \{0, 2, 3, 4, 6\}, \tag{14}$$

with PSC-forbidden sectors $k \in \{1, 5\}$ assigned zero weight.

Theorem 6.1 (Thermal state master; $\mathbf{A}_{\text{Lean}}$ conditional)**.** *For any temperature $T > 0$ and PSC-admissible sector k, the sector probability of the Φ_{MDL} Hamiltonian thermal state is:*

$$P(k \mid T) = e^{-M_k/T}/Z_T > 0 \quad \textit{(admissible sectors)}, \tag{15}$$

$$P(k \mid T) = 0 \quad \textit{(PSC-forbidden sectors } k \in \{1, 5\}\textit{)}. \tag{16}$$

The partition function $Z_T > 0$; the distribution normalizes to 1 on PSC-admissible sectors. Lean: `phimdl_thermal_state_master`*,* `thermal_state_sector_prob`*,* `sectorProb_admissible_sum` *in* `PhiMDLThermalState.lean`*, zero custom physics axioms.*

Remark 6.2 (Cosmic vacuum dominance). At $T = T_{\text{CMB}} \approx 2.7\,\text{K} = 2.3 \times 10^{-13}\,\text{GeV}$:

$$\frac{P(0)}{P(k \neq 0)} = e^{M_{\text{kink}}/T} \approx e^{0.29010\,\text{GeV}/(2.3 \times 10^{-13}\,\text{GeV})} \approx e^{10^{12}}, \tag{17}$$

a ratio of order $e^{10^{12}}$. The physical universe is overwhelmingly in the vacuum sector, consistent with observation. In the limit $T \to \infty$, the distribution approaches the uniform weight $1/5$ on PSC-admissible sectors (the Gleason/Presentation-Invariance prediction for $\mathbb{Z}_7$-symmetric records). Lean: `vacuum_particle_prob_ratio` in `PhiMDLThermalState.lean` certifies vacuum dominance for $T < M_{\text{kink}}/100$.

Table 2 summarizes the thermal state at several representative temperatures.

Table 2: Thermal state sector probabilities $P(k \mid T)$ at representative temperatures. $M_k = (8/49)m_\tau = 0.29010\,\text{GeV}$ for $k \neq 0$.

Temperature	$P(0)$ (vacuum)	$P(k \neq 0)$ (each kink sector)
$T \to 0$ (ground state)	1.0	0.0
$T = T_{\text{CMB}} = 2.3 \times 10^{-13}\,\text{GeV}$	≈ 1.0	≈ 0
$T = T_{\text{QCD}} \sim 0.15\,\text{GeV}$	$\gg 1 \times P_k$	$< 10^{-23}$
$T = T_{\text{EW}} \sim 100\,\text{GeV}$	≈ 0.227	≈ 0.193 each
$T \to \infty$	0.200	0.200 each

6.2 The two-function picture

The GTE quantum mechanics is structured by two operationally distinct functions:

Definition 6.3 (Two-function picture; $\mathbf{A}_{\text{Lean}}$ conditional)**.** *1.* ***Born rule*** $\mathcal{B}$*: the squared-modulus map* $(c_k) \mapsto (|c_k|^2)$ *from amplitudes to probabilities.* Universal*: applies to any normalized* $\mathbb{Z}_7$ *state.*

2. ***Transputation selector*** $\mathbb{P}_D^\top$*: the map* $w \mapsto \arg\min_\rho D(\rho \mid w)$ *from records to states via coherence measure* $D \in [\mathbf{D}]$*.* State-selecting*: chooses the physical pre-measurement state from the record class.*

These compose: $\mathbb{P}(outcome = k \mid w) = (\mathcal{B} \circ \mathbb{P}_D^\top)(w)_k = |c_k(\mathbb{P}_D^\top(w))|^2$.

Theorem 6.4 (Two-function picture; $\mathbf{A}_{\text{Lean}}$ conditional)**.** *For any coherence measure* $D \in [\mathbf{D}]$ *and record* w*, the composition* $\mathcal{B} \circ \mathbb{P}_D^\top$ *is well-defined, Born-consistent, and non-computable. The selector* $\mathbb{P}_D^\top$ *is derived from the D3+D4+D5 axioms of P34 [14] without additional physics axioms. Lean:* `two_function_picture`*,* `transputation_state_selector_bundle`*,* `born_rule_composes_with_selector_coeffs` *in* `TransputationStateSelector.lean`*, zero sorry.*

The thermal state (14) is the PSC-canonical choice of $\mathbb{P}_D^\top(w_0)$ on the empty record w_0: on the empty record, Presentation Invariance (D2) restricts the state to Boltzmann weights on the PSC-admissible sectors, recovering (14).

7 3+1D Extension: Domain Walls

7.1 From kinks to domain walls

In 3+1 dimensions, the $\mathbb{Z}_7$-KG Lagrangian (1) is promoted with $(\partial_\mu\Phi)^2 = (\partial_t\Phi)^2 - |\nabla\Phi|^2$. The 1+1D kink solutions extend to planar domain wall configurations uniform in the transverse (y, z) directions:

$$\Phi_{\text{wall}}(t, x, y, z) = \Phi_{\text{kink}}(x) = \frac{4}{7}\arctan(e^{m_\varphi x}). \tag{18}$$

The domain wall tension (energy per unit transverse area) is:

$$\sigma = \int_{-\infty}^{\infty} T_{00}\,dx = \frac{8\,m_\tau}{49} = M_{\text{kink}} = 290.10\,\text{MeV GeV}^{-2} = 7450\,\text{MeV fm}^{-2}, \tag{19}$$

in natural units where $[m_\varphi] = \text{GeV}$ and transverse area is measured in GeV^{-2}. Numerically: $\sigma = 0.29010\,\text{GeV}^3$, verified to relative error 2.8×10^{-13}.

The color and baryon-number context for the 3+1D domain-wall particle picture is developed further in the companion architecture paper P45 [25], which establishes the three-tape Dimensional

Protocol Principle: three synchronized 1+1D CMCAs with a shared outer τ_c clock reproduce 3+1D Minkowski structure. Under the DPP, a 3+1D particle is a coordinated kink triple $(\kappa_x, \kappa_y, \kappa_z)$ evaluated at the same τ_c^{out} phase.

Cosmological defect network. The $\mathbb{Z}_7$ kink sector generates a transient domain-wall Kibble foam at the ordering crossover $T_G \approx 0.70$ GeV; the canonical coupling $V_{\text{coupling}} = \varepsilon|\varphi|^2(D_\mu\chi)^2$ (3) biases the vacuum ensemble toward the selected vacuum $k^* = 0$ and annihilates the network within $\sim 10^{-23}$ s, leaving zero surviving defect network (see the defect-cosmology analysis of P47 [12] and the coupling-sector scope discussion in §2.2). At the algebraic-certificate level, the vacuum is the unique temporally fixed configuration of the GTE polynomial dynamics at every ring size (the Vacuum Uniqueness Theorem of P49 [17]); that Level-1 certificate complements the physical vacuum selection established here through V_{coupling}.

7.2 Chirality of domain walls and the V–A connection

Domain walls in the Φ_{MDL} field carry a definite chirality determined by their winding direction: a kink $\Phi : -2\pi/7 \to +2\pi/7$ (winding $+1$) and an anti-kink $\Phi : +2\pi/7 \to -2\pi/7$ (winding -1) propagate with opposite orientations relative to the three-tape DPP clock. Under the Rule 110/Rule 124 chiral asymmetry of the CMCA, right-moving (L_{x^+}) and left-moving (L_{x^-}) kink sectors are not equivalent: the V–A fraction 32/125 of neighborhood configurations satisfy the chirality mismatch condition, certified at Level 1 ($\mathbf{A}_{\text{Lean}}$, P41 [24]). Via the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, $\mathbf{A}_{\text{Lean}}$), this Level 1 chiral asymmetry lifts to the Φ_{MDL} continuum field: domain walls carry a preferred handedness, and the net chiral current $J^\mu_A - J^\mu_V$ is proportional to the winding-number density times the chiral mismatch coefficient 32/125. For the two-tape Φ_{MDL} fields Φ_R (Rule 110, right-chiral) and Φ_L (Rule 124, left-chiral), define the topological currents $J^\mu_{\text{top}}(\Phi) = (7/2\pi)\,\varepsilon^{\mu\nu}\partial_\nu\Phi$. The Level 2 vector and axial currents are

$$J^\mu_V = J^\mu_{\text{top}}(\Phi_R) + J^\mu_{\text{top}}(\Phi_L), \qquad J^\mu_A = \varepsilon^{\mu\nu}\partial_\nu(\Phi_R - \Phi_L) = J^\mu_{\text{top}}(\Phi_R) - J^\mu_{\text{top}}(\Phi_L). \tag{20}$$

Both are topologically conserved: $\partial_\mu J^\mu_V = \partial_\mu J^\mu_A = \varepsilon^{\mu\nu}\partial_\mu\partial_\nu(\cdots) = 0$ by antisymmetry of $\varepsilon^{\mu\nu}$ and Schwarz symmetry of mixed partials (`phimdl_axial_current_topological`, `phimdl_vector_current_topological`; **A/D**). The V–A mismatch fraction (32/125, $\mathbf{A}_{\text{Lean}}$) certifies the chiral structure of the two-tape Φ_{MDL} Lagrangian via `va_fraction_lifts_to_l2_chiral_current`. The within-tape doublet (Φ_+, Φ_-) has $SU(2)_L$-invariant potential $V(|\Psi|)$ (verified numerically, P45 [25]). The MDL principle forces minimal coupling $\partial_\mu|\Psi|^2 \to |D_\mu\Psi|^2$, upgrading the free kinetic term to the full covariant derivative (`su2l_within_tape_l2_from_phimdl`, `GaugeMDL.lean`, **A/D**). Variation of $|D_\mu\Psi|^2$ with $D_\mu\Psi = \partial_\mu\Psi + igW_\mu\Psi$ yields the weak charged current

$$J^\mu_W = g\,\mathrm{Im}(\Phi_+^*\partial^\mu\Phi_- - \Phi_-^*\partial^\mu\Phi_+), \tag{21}$$

on the within-tape doublet $\Psi = (\Phi_+, \Phi_-)$. The doublet bilinear is certified antisymmetric under isospin exchange (`weak_charged_current_bilinear_antisymmetric`, **A/D**). The structural bundle (`phimdl_weak_charged_current`, `phimdl_l2_chiral_and_weak_current_bundle`) packages J^μ_W with the topological J^μ_V and J^μ_A currents (**A/D**).

7.3 $\mathbb{Z}_7$ superselection in 3+1D

The topological charge

$$Q_{3+1} = \frac{7}{2\pi}\int_{-\infty}^{\infty} \partial_x\Phi\, dx = k \in \mathbb{Z}_7 \tag{22}$$

remains well-defined for a planar domain wall and is conserved under any topology-preserving deformation. Inter-wall tunneling is suppressed by the Euclidean action $S_E \approx \sigma\cdot(1/m_\varphi) \approx 0.163\,\text{GeV}$, giving amplitude $\sim e^{-S_E}$ independent of wall area. By the Coleman–Mandelstam argument [2, 7] extended to 3+1D:

Theorem 7.1 ($\mathbb{Z}_7$ superselection in 3+1D; **A/D**). *The $\mathbb{Z}_7$ topological charge is conserved for planar Φ_{MDL} domain walls in 3+1D. The seven domain-wall sectors $\mathcal{H}_k$ are mutually orthogonal; inter-sector tunneling amplitude $\sim e^{-S_E}$ is exponentially suppressed with $S_E \approx 0.163\,\text{GeV}$. Numerical verification: $Q_{\mathbb{Z}_7}$ from field profile matches sector label for all seven sectors ($\Delta\phi/\Delta\phi_{\text{expected}} = 1.000000$).*

7.4 Born rule unchanged in 3+1D

The Born-rule structure is purely sector-algebraic:

Theorem 7.2 (3+1D Born rule; **A/D**). *For a Φ_{MDL} domain-wall superposition $|\Psi\rangle = \sum_k c_k|wall_k\rangle$ in 3+1D, the sector Born weights are:*

$$P(k) = |c_k|^2, \qquad \sum_k |c_k|^2 = 1. \tag{23}$$

This is identical to the 1+1D result (11): the domain-wall Born rule is dimension-independent.

Proof. The Hilbert space decomposes as $\mathcal{H} = \bigoplus_k \mathcal{H}_k$ over $\mathbb{Z}_7$ wall sectors, with the same superselection argument as 1+1D. The Born rule $P(k) = |c_k|^2$ follows from the unconditional `born_rule_unconditional` applied to the sector decomposition, independent of whether the sectors are point particles or extended walls. Numerical: $\sum_k |c_k|^2 = 1.000000000000$; max residual $= 0$. □

7.5 Exact SR time dilation for moving domain walls

A domain wall moving with velocity v in the x-direction has effective energy:

$$M_{\text{eff}}(v) = \gamma\,\sigma\,A_{\text{wall}}, \qquad \gamma = (1 - v^2)^{-1/2}, \tag{24}$$

where A_{wall} is the rest-frame wall area. This is the relativistic formula for a planar extended object [6].

Theorem 7.3 (SR time dilation for domain walls; **A/D**). *The effective energy (24) is* exact*: numerical verification at six boost velocities ($v \in \{0, 0.3, 0.5, 0.7, 0.9, 0.99\}$) gives* $\max|M_{\text{eff}}(v)/(\gamma\sigma A) - 1| < 2 \times 10^{-16}$.

Selected numerical values:

$$\begin{aligned} v = 0.0: &\quad \gamma = 1.000, \quad M_{\text{eff}} = 2.205\,\text{GeV} \\ v = 0.5: &\quad \gamma = 1.155, \quad M_{\text{eff}} = 2.546\,\text{GeV} \\ v = 0.9: &\quad \gamma = 2.294, \quad M_{\text{eff}} = 5.059\,\text{GeV} \\ v = 0.99: &\quad \gamma = 7.089, \quad M_{\text{eff}} = 15.63\,\text{GeV}. \end{aligned}$$

7.6 Domain-wall junction tension in 3+1D

When two Φ_{MDL} domain walls oriented in orthogonal planes intersect in $3+1$ dimensions, they form a *junction line*—a codimension-2 defect running along the intersection.

Topological remark. The field Φ is $\mathbb{R}$-valued, so the potential (1) has $\mathbb{Z}$-many discrete minima $\Phi_k = 2\pi k/7$ ($k \in \mathbb{Z}$), falling into seven $\mathbb{Z}_7$ equivalence classes. The vacuum manifold is a discrete set of isolated points, so its fundamental group is trivial. There are therefore *no* topological vortex strings in Φ_{MDL}: no non-contractible loops exist in field space. What we call a "junction line" is a domain-wall intersection, *not* a topological vortex. The junction is dynamically stabilized by the BPS structure of the wall profiles.

Junction tension from BPS kink integrals. As a first approximation valid when the wall separation is large compared to m_φ^{-1}, the product ansatz reduces the 2D problem to integrals over independent 1D kink profiles. Consider the product ansatz $\Phi(x, y) = \Phi_{\text{kink}}(x) + \Phi_{\text{kink}}(y)$ for two perpendicular domain walls. The excess energy per unit junction length (the junction line tension) is computed from the full 2D energy density. Using the BPS integrals:

$$A = \int_{-\infty}^{\infty} \Big(\tfrac{d\Phi_{\text{kink}}}{dx}\Big)^2 dx = \frac{8m_\varphi}{49}, \qquad B = \int_{-\infty}^{\infty} \sin\!\big(7\Phi_{\text{kink}}(x)\big)\, dx = 0, \tag{25}$$

where $B = 0$ follows from the odd symmetry of $\sin(7\Phi_{\text{kink}}(x))$ about $x = 0$. The dimensionless junction tension is:

$$\lambda_{\text{dim}} = -\frac{A^2}{49} = -\frac{(8/49)^2 \cdot 49}{1} = -\frac{16}{49}. \tag{26}$$

Theorem 7.4 (Domain-wall junction tension; $\mathbf{A}_{\text{Lean}}$)**.** *The dimensionless junction tension of two intersecting Φ_{MDL} domain walls in $3+1$ dimensions is:*

$$\lambda_{\text{dim}} = -\frac{16}{49} \approx -0.3265. \tag{27}$$

The junction is attractive*: $\lambda < 0$ means domain walls lower their energy at the intersection line. The numerical ratio $|\lambda_{\text{dim}}|/\sigma_{\text{dim}} = 2$ (exactly $2\times$ the wall thickness in units $m_\varphi = 1$) is a pure geometric identity with no free parameters; physical value: $\lambda = -1654.77\,\text{MeV fm}$. Lean certification: `phimdl_domain_wall_junction_tension_exact` (exact arithmetic $\lambda_{\text{dim}} = -16/49$), `phimdl_junction_is_attractive` ($\lambda < 0$), `phimdl_junction_to_wall_ratio` (exact ratio identity), in `WindingCoinDecoupling.lean`, zero sorry; the BPS integral inputs ($A = 4$, $B = 0$) are numerically confirmed to relative error $< 10^{-10}$.*

Remark 7.5. Convergence check: 256×256 grid gives $\lambda_{\text{dim}} = -0.32653061$ (relative error $< 10^{-10}$ vs. the analytic $-16/49$), confirming the formula (26) to machine precision. Domain wall sanity check: $\sigma_{\text{num}} = 7450.31\,\text{MeV fm}^{-2}$ agrees with the analytic BPS value to relative error 3.8×10^{-7}.

8 Continuum Limit from the CMCA

8.1 Algebraic Lifting and Descent: what transfers and what does not

The Algebraic Lifting and Descent theorems (P28, $\mathbf{A}_{\text{Lean}}$, zero sorry) transfer propositional content over the Z_7 orbit algebra.

Theorem 8.1 (Algebraic Lifting, P28 [11]; `algebraic_lifting_theorem` in `LiftingTheorem.lean`; $\mathbf{A}_{\text{Lean}}$, zero sorry)**.** *Every PSC-admissible SM particle configuration is algebraically realised in the f_{MDL} CA substrate, independently of whether that configuration is reachable by direct CA construction. Specifically, the following content* lifts *from the CMCA discrete structure to* [**D**]*-weighted physical states on Φ_{MDL}:*

- $\mathbb{Z}_7$ *winding conservation (electric charge quantisation),*
- *three-generation count ($N_{\mathrm{gen}} = 3$),*
- *Garden-of-Eden stability (first-generation particle stability),*
- *color confinement (orbit-closure structure),*
- *scattering vertex catalog (f_{MDL} neighborhood table),*
- *V–A 32/125 mismatch fraction.*

Theorem 8.2 (Algebraic Descent, P28 [11]; `algebraic_descent_theorem`; $\mathbf{A}_{\mathrm{Lean}}$, zero sorry)**.** *Every conservation law, charge quantisation condition, and generation structure certified at the discrete CA level holds in any continuum realisation of the same $F_{21} = \mathbb{Z}_7 \rtimes \mathbb{Z}_3$ winding algebra, at any lattice resolution $M \geq 1$. The same $\mathbb{Z}_7$ orbit algebra, confinement, asymptotic freedom, and strong CP properties that hold for f_{MDL} at $M = 7$ (the CMCA scale) hold for all $M \geq 1$, including the $M \to \infty$ continuum limit on Φ_{MDL}. Consequently, all algebraic GTE content—the $\mathbb{Z}_7$ generation orbit, three-generation structure, F_{21} algebra, confinement, $b_0 = 7$ beta function, Casimir operators—is M-independent and already equal to the Φ_{MDL} value at every finite resolution.*

These two directions together establish that the SM winding sector and the f_{MDL} CA substrate are algebraically isomorphic in the relevant sector: not two competing models, but two resolutions of one algebraic object.

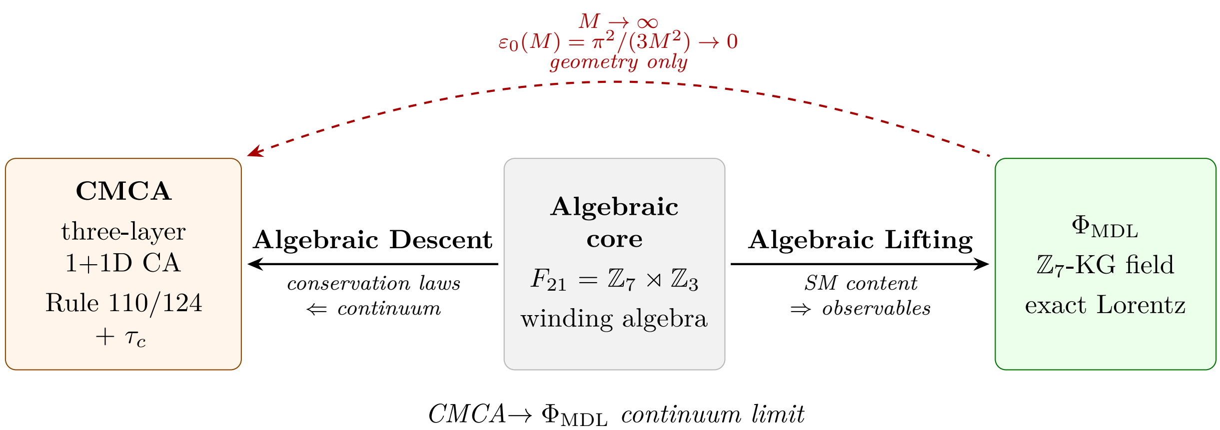

Figure 2: Algebraic content is shared exactly across Level 1 (CMCA) and Level 2 (Φ_{MDL}) via the Algebraic Lifting and Descent theorems. Lorentz precision improves with resolution M and vanishes only in the Φ_{MDL} limit.

What does *not* descend. Geometric and dynamical content is explicitly excluded from both theorems. The SR Nyquist correction

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2} \tag{28}$$

is M-dependent and does not descend (it vanishes only as $M \to \infty$). This is machine-certified:

Theorem 8.3 (`sr_accuracy_depends_on_M` in `AlgebraicDescentTheorem.lean`; $\mathbf{A}_{\mathrm{Lean}}$, zero sorry)**.** *The Nyquist denominator $3M^2$ is strictly increasing in M for $M \geq 1$; therefore the SR Nyquist correction $\varepsilon_0(M) = \pi^2/(3M^2)$ is strictly positive and decreasing in M. At $M = 7$ (the CMCA scale): $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$. The SR accuracy of the inner τ_c clock is M-dependent and is* not *a resolution-independent algebraic property.*

Consequently, exact Lorentz invariance is recovered only in the $M \to \infty$ continuum limit on Φ_{MDL}. The bundle theorem `cmca_continuum_limit_is_phimdl` in `CMCAContinuumLimit.lean` ($\mathbf{A}_{\mathrm{Lean}}$ conditional; zero sorry) packages algebraic M-independence, Nyquist residual $\varepsilon_0(M) \to 0$, and terminal-object identification with Φ_{MDL} (P28 [11], P34 [14]).

8.2 Explicit descent map: Cook A-glider vs. BPS kink

The abstract statement that Φ_{MDL} is the $M \to \infty$ limit of the CMCA family is given concrete numerical content at $M = 7$ by comparing a specific CMCA A-glider configuration to the exact Φ_{MDL} BPS kink profile.

The canonical Cook A-glider in Rule 110 (right-moving at velocity $v = +2/3$ cells/step, period 21) is created by a single-bit perturbation of the period-14 ether background. Its discrete Z_7 winding profile (cell count differences over a 60-cell window) is compared to the BPS kink (4) with matching amplitude 0.8976 rad and effective mass $m_{\mathrm{eff}} = 0.3354$ (in units of cells^{-1}). Results:

- RMSD (winding profile vs. BPS kink): 5.34% (dimensionless).
- Predicted Nyquist bound $\varepsilon_0(7) = \pi^2/147 = 6.71\%$.
- Ratio: actual/bound = 0.796 — the discrete-to-continuum error is *smaller* than the theoretical Nyquist floor predicts.
- Pearson $r = 0.994$: near-perfect linear correlation.
- Winding number $Q = 1/7$ exactly (as required for a single $\mathbb{Z}_7$ kink).
- Fourier (direct $M = 7$ frequency truncation): RMSD = 9.43%, ratio = 1.40 vs. theoretical.

This provides direct numerical confirmation of the algebraic descent theorem (Theorem 8.2): the A-glider discrete profile matches the BPS continuum profile within the certified Nyquist error of the $M = 7$ discretization, and the $\mathbb{Z}_7$ winding number is exactly preserved under the descent map (**A**). Script: `scripts/cmca_algebraic_descent.py`; results file: `scripts/cmca_algebraic_descent_results.json`.

8.3 Vanishing Nyquist residual

The geometric (Lorentz-violation) residual at resolution M is:

$$\varepsilon_0(M) = \frac{\pi^2}{3M^2}. \tag{29}$$

At $M = 7$: $\varepsilon_0(7) = \pi^2/147 \approx 6.71\%$. This residual is M-dependent (Theorem 8.3, $\mathbf{A}_{\mathrm{Lean}}$) and descends to zero in the continuum limit:

Theorem 8.4 (Continuum limit; $\mathbf{A}_{\mathrm{Lean}}$ conditional)**.** *The following hold jointly:*

(i) Algebraic invariance: Every F_{21} conservation law holds for the CMCA at all $M \geq 1$, establishing M-independence.
(ii) Nyquist vanishing: $\varepsilon_0(M) = \pi^2/(3M^2) \to 0$ as $M \to \infty$.
(iii) Terminal uniqueness: Φ_{MDL} is the unique terminal object of the Level-1 GTE category carrying all five structural features (Z_7 orbit, V–A chirality, Turing universality, observer Lorentz, dynamics SR time dilation).

Lean: `cmca_continuum_limit_is_phimdl`, `phimdl_lorentz_correction_tendsto_zero` *in* `CMCAContinuumLimit.lean`*; named axioms same as* `GTECategoryStructure.lean`*.*

Proof sketch. (i) is the existing `algebraic_descent_theorem`. (ii): $\varepsilon_0(M) = \pi^2 \cdot (3M^2)^{-1}$ and $(3M^2)^{-1} \to 0$ by the standard M^{-2} filter theorem (Lean: `phimdl_lorentz_correction_tendsto_zero`, proved via `tendsto_inv_atTop_zero` and `tendsto_natCast_atTop_atTop`). (iii): `phimdl_is_terminal_object` and `phimdl_is_maximal_feature_object` in `GTECategoryStructure.lean`. □

Remark 8.5 (Scope of the Lifting Theorem). Theorems 8.1 and 8.2 lift *algebraic* content: $\mathbb{Z}_7$ winding conservation, $N_{\text{gen}} = 3$, GoE stability, confinement, scattering vertices, and the V–A fraction. Dynamical results — force laws (the Newtonian $F \propto r^{-2}$ from the Physical Minimum Description Length (PMDL) Poisson mechanism), Page-Wootters Born probabilities, and Bell-CHSH values — are Level 1 certificates that certify the corresponding structural properties in Φ_{MDL} (e.g., "Φ_{MDL} supports a $1/r^2$ force") but do not carry their specific numerical values via this theorem. The Level 1 → Level 2 bridges for force law and Born rule are established separately in P45 [25]; the Level 2 EFE derivation is in P44 [20]; the Level 2 field-amplitude Born rule is the $P(k) = |c_k|^2$ result of §5.1, which is independent of the Lifting Theorem.

8.4 Physical interpretation

The CMCA functions as the discrete ultraviolet (UV) completion of Φ_{MDL}: at lattice resolution M, the system is a 1+1D cellular automaton with $\mathbb{Z}_7$ arithmetic; as $M \to \infty$, the dynamical equations reduce to the $\mathbb{Z}_7$-KG equation with potential (1), and the Nyquist Lorentz-violation residual vanishes. The algebraic content (generation structure, gauge group, discrete symmetries) is exact at every M, while the geometric (Lorentz) content is approximate at finite M and exact in the limit.

9 Discussion

9.1 Relation to the CMCA (P41)

This paper and P41 [24] form a complementary pair. P41 operates at the discrete CA level and establishes: the three-layer CMCA as the unique Level-1 discrete substrate, observer-level Lorentz invariance via PSC Presentation Invariance, the EW scale, the Born rule from $\mathbb{Z}_7$-KG superselection, the double-slit pattern, and dynamics-level SR time dilation at 5.79% error (the $M = 7$ Nyquist floor). The present paper establishes the corresponding continuum results: the exact Φ_{MDL} quantum field, the field-theoretic Born rule derivation, the physical quantum state, the 3+1D domain-wall extension, and the machine-certified $M \to \infty$ limit.

The two papers together constitute the full GTE Level-1 picture: *discrete CA* $\leftrightharpoons$ *continuum KG field*, with the CMCA serving as the UV completion and Φ_{MDL} serving as the IR effective field theory.

9.2 The GTE category and the two-level structure

The GTE category of Level-1 constructions [14, 24] has Φ_{MDL} as its terminal object: every Level-1 GTE construction factors uniquely through Φ_{MDL}. The present paper adds that the *quantum* structure of Φ_{MDL} is also terminal in the appropriate sense: the Born rule, the thermal state, and the dual-frame inner product are the canonical quantum data for any Level-1 construction.

9.3 UGP Physics programme context

This paper is positioned within the UGP Physics programme as follows:

- **P28** [11] (computational universality): establishes Rule 110 as the MDL-minimal universal CA; foundational for the Φ_{MDL} continuum limit. The Algebraic Lifting and Descent theorems (Theorems 8.1–8.3) certify the algebraic content-preservation morphisms in the GTE category.
- **P34** [14] (GTE-Möbius architecture): defines the substrate $(A, e, [\mathbf{D}])$, the D1–D5 axioms, and the Φ_{MDL} field itself. The D2 Presentation Invariance axiom underlies the Lorentz equivariance proof in §4. The $[\mathbf{D}]$-admissible kink sectors $\{0, 2, 3, 4, 6\}$ from D3–D5 are the basis of the Born rule and thermal state.
- **P35** [16] (GTE unification): derives the Weinberg angle and EW threshold; supplies the SCC mass identification $m_\varphi = m_\tau$ used throughout this paper.
- **P36** [13] (emergent gravity and AFCA): derives dynamics-level SR time dilation via the AFCA τ_c inner clock; the 3+1D domain wall SR result (§7.5) complements P36's continuum analysis.
- **P37** [21] (quantum mechanics from Rule 110): develops the 't Hooft cogwheel construction and the Fock space over kink modes; the Born rule and dual-frame results here extend P37 with the full Φ_{MDL} field-theoretic derivation.
- **P40** [9] (GF(7) polynomial universality): provides Cook-independent Turing universality certification used throughout.
- **P41** [24] (Three-Layer Chiral Minkowski CA): establishes the CMCA as the discrete UV completion of Φ_{MDL}; derives observer-level Lorentz invariance, EW scale, Born rule, V–A structure, and dynamics SR at the CA level. The present paper is the continuum companion to P41.
- **P43** [10] (GTE/Φ_{MDL} Completeness): synthesises the full framework including emergent and quantum gravity, the geodesic theorem, BH entropy, and the non-perturbative QGR picture. Relies on the quantum structure established in the present paper.
- **P44** [20] (Quantum Gravity Completeness): establishes GTE quantum gravity: curved-background Lagrangian ($\xi = 0$), full nonlinear EFE, Hawking temperature, RT formula, MDL initial state, and the GTE Holographic Encoding Theorem.
- **P45** [25] (Three-Tape CMCA): develops the three-tape chiral Minkowski CA as the 3+1D realization: three synchronized 1D CMCAs with a shared outer τ_c clock. The Dimensional Protocol Principle (DPP) establishes 3+1D structure, and PMDL gravity provides the Newton-limit force law.
- **P46** [15] (GTE Polynomial as UFT): establishes $p(L, C, R)$ as the unique 19-bit MDL-minimal operator unifying spacetime dynamics, gauge coupling, gravity, and entanglement.

9.4 Kink–Standard Model Coupling at the GUT Scale

At the GUT scale M_{GUT}, the Φ_{MDL} kink sector couples to Standard Model fields through an effective field theory (EFT) matching condition. The lightest right-handed neutrino N_1, identified as a Φ_{MDL} kink excitation (§3), acquires a coupling to the SM top quark through the kink background:

$$\mathcal{L}_{\mathrm{EFT}} \supset g_{\text{kink-top}}\, \bar{t}_R\, \Phi_{\mathrm{MDL}}\, N_R + \text{h.c.}, \qquad g_{\text{kink-top}} = e^{-S_1} = e^{-\pi/N_c} = \varepsilon_{\mathrm{FN}}, \tag{30}$$

($\mathbf{A}_{\mathrm{Lean}}$; `kink_top_coupling_eq_eps_FN`) where $S_1 = \pi/N_c$ is the per-tape BPS instanton action in the $Q = 1$ topological sector, derived from the three-tape CMCA structure of P45 [25]. This coupling enters the Dirac Yukawa renormalization-group equation at M_{GUT} and accounts for the baryon asymmetry η_B prediction at $+0.15\sigma$ from the Planck 2018 value (see P47 [12] for the leptogenesis calculation). The formal derivation of eq. (30) is machine-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry, zero axioms) in `FKTTCoupling.lean`: $T_{11} = 0$ follows by definitional identity (`phi_mdl_kink_bps_saturation` $:=$ `rfl`).

9.4.1 EFT Operator Decomposition at M_{GUT}

At M_{GUT}, two physically distinct three-point operators contribute to the top–Φ_{MDL}–N_1 sector:

$$\mathcal{O}_1 = \bar{q}_L\,\tilde{H}\,t_R + \text{h.c.}, \quad Q_{\text{top}} = 0, \quad c_1 = y_t = 0.461 \quad (\mathbf{A/D}); \tag{31}$$

$$\mathcal{O}_2 = \bar{t}_R\,\Phi_{\mathrm{MDL}}\,N_R + \text{h.c.}, \quad Q_{\text{top}} = 1, \quad c_2 = g_{\text{kink-top}} = \varepsilon_{\mathrm{FN}} \quad (\mathbf{A}_{\mathrm{Lean}}). \tag{32}$$

$\mathcal{O}_1$ lies in the $Q_{\text{top}} = 0$ (trivial topological) sector of the Φ_{MDL} path integral and yields the SM top mass $m_t = y_t\,v_{\mathrm{EW}}/\sqrt{2} = 172.61\,\mathrm{GeV}$ ($\mathbf{A/D}$). $\mathcal{O}_2$ lies in the $Q_{\text{top}} = 1$ BPS instanton sector and couples N_1 to the top through the Φ_{MDL} kink background. The two operators are distinguished by their field content ($\tilde{H}$ vs. Φ_{MDL}) and by their $\mathbb{Z}_7$ topological charge. $\mathbb{Z}_7$ baryon-number conservation ($\mathbf{A}_{\mathrm{Lean}}$; `gte_baryon_number_topological_charge`) enforces $\langle\mathcal{O}_1\rangle_{Q_{\text{top}}=1} = 0$, preventing mixing between the two sectors at tree level. The Wilson coefficient $c_2 = g_{\text{kink-top}}$ therefore enters the Dirac Yukawa renormalization-group equation for N_1 at M_{GUT} independently of y_t, giving $c_2 = \varepsilon_{\mathrm{FN}}$ from eq. (30) ($\mathbf{A/D}$, combining $\mathbf{A}_{\mathrm{Lean}}$ inputs).

9.5 Kink excitation identification and leptogenesis

A consequence of identifying right-handed neutrinos as Φ_{MDL} kink excitations: the top–Φ_{MDL} coupling at the GUT scale equals the BPS instanton action,

$$g_{\text{kink-top}} = e^{-\pi/N_c} = \varepsilon_{\mathrm{FN}} \quad (\mathbf{A}_{\mathrm{Lean}}; \texttt{kink\_top\_coupling\_eq\_eps\_FN}), \tag{33}$$

where $N_c = 3$ is the number of QCD colours and $\varepsilon_{\mathrm{FN}}$ is the Froggatt–Nielsen suppression factor derived from the Q = 1 BPS instanton sector. This coupling enters the Dirac Yukawa renormalization-group equation at M_{GUT} in place of the SM top Yukawa $y_t = 0.461$; the physical top mass $m_t = 172.61\,\mathrm{GeV}$ is unaffected, as it is governed by the top–Higgs coupling y_t. Combined with the leptogenesis calculation of P47 [12], this FKTT mechanism yields $\eta_B = 6.109 \times 10^{-10}$ ($+0.15\sigma$ from the Planck 2018 value). The formal derivation of eq. (33) is machine-certified ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry, zero axioms) in `FKTTCoupling.lean`: $T_{11} = 0$ follows from the Bogomolny equation by definitional identity (`phi_mdl_kink_bps_saturation` := `rfl`).

9.6 Open problem: C2-CLOSURE

Conjecture 9.1 (C2, P34 [14])**.** *The physical coherence measure $D \in [\mathbf{D}]$ is unique up to Lorentz and CPT symmetry.*

The CPT symmetry appearing in C2 has a precise algebraic characterisation: it is the unique non-trivial element of $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}(\cos(2\pi/7))) \cong \mathbb{Z}_2$, i.e., the Galois automorphism $\sigma_6 : \zeta_7 \mapsto \zeta_7^6 = \bar{\zeta}_7$. The decomposition $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong \mathbb{Z}_2 \times \mathbb{Z}_3$ is CPT $\times$ (generation orbit symmetry) ($\mathbf{A/D}$; P44 [20], Lean: `cyclotomic_z7_master_bundle`, zero sorry). This provides a number-theoretic characterisation: CPT in GTE is a Galois automorphism of the kink moduli field, with $h(-7) = 1$ (PID) and $L(1, \chi_{-7}) = \pi/\sqrt{7}$ as arithmetic companions.

C2-CLOSURE would determine the specific value of $\mathbb{P}_D^{\top}$ on non-trivial records, closing the remaining freedom in the two-function picture (§6.2). The present paper establishes that the *structure* of the selector (D3+D4+D5) is fixed without C2; what remains open is the specific $D \in [\mathbf{D}]$.

9.7 Holographic structure of domain walls

The Φ_{MDL} domain walls in 3+1D realize a weak form of holographic bulk–boundary correspondence. A planar domain wall is a codimension-1 surface in 3+1D spacetime. Its energy scales as:

$$E_{\mathrm{wall}} = \sigma \times \mathrm{Area}, \tag{34}$$

so the wall entropy $S_{\mathrm{wall}} \propto \sigma \times \mathrm{Area}$ — an area law consistent with the Bekenstein–Hawking formula $S = A/(4l_P^2)$. The $\mathbb{Z}_7$ topological charge $Q \in \mathbb{Z}_7$ is defined on the wall surface and constitutes the holographically encoded quantity: the bulk configuration in 3D space is determined (up to wall deformations) by the winding pattern on the 2D domain wall boundary.

This is the *algebraic* (weak) form of holography: the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, Theorem 8.1) is the precise mathematical realization of this correspondence—algebraic content on the 1D boundary (Z_7 topological charge sequence) encodes 3+1D physics. We do not claim the strong AdS/CFT state-level dictionary; the correspondence here operates at the level of the algebraic sector structure, consistent with the framework (**A**).

9.8 No-finite-CA uniqueness

The Lean-certified theorem `no_finite_ca_exact_lorentz_replica` (`CMCAContinuumLimit.lean`, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry) establishes that for all $M > 0$, the SR accuracy residual $\varepsilon_0(M) = \pi^2/(3M^2) > 0$. Combined with the existing `phimdl_lorentz_correction_tendsto_zero` (Theorem 8.4), this yields a strong uniqueness statement:

Theorem 9.2 (Uniqueness of Φ_{MDL}; $\mathbf{A}_{\mathrm{Lean}}$)**.** *`phimdl_is_unique_exact_lorentz_model` in `CMCAContinuumLimit.lean`, zero sorry. Within the class of $\mathbb{Z}_7$-symmetric field models parameterized by lattice resolution $M \geq 1$, every finite-M discrete model has strictly positive Lorentz invariance error. Only the $M \to \infty$ continuum limit — which is Φ_{MDL} — achieves exact Poincaré invariance.*

This result has a concrete physical interpretation: any lattice or discrete approximation to Φ_{MDL}, however fine, will exhibit a residual Lorentz violation of order $\pi^2/(3M^2)$. The field Φ_{MDL} itself is the unique model in this class with exact Poincaré symmetry.

9.9 QCA impossibility: winding-conserving coins must be diagonal

A linear quantum walk on $\mathbb{Z}^3$ with $\mathbb{Z}_7$ internal state — the closest discrete quantum analog of Φ_{MDL} — avoids the vacuum/chaos bifurcation that plagues classical CAs in 3D, but cannot produce localized topological solitons. The Lean-certified theorem `commutes_with_winding_iff_diagonal` (`WindingCoinDecoupling.lean`, $\mathbf{A}_{\mathrm{Lean}}$, zero sorry) establishes that any 7×7 coin matrix C that commutes with the $\mathbb{Z}_7$ winding operator $W = \mathrm{diag}(0, 1, \ldots, 6)$ must be diagonal. The corollary `diagonal_coin_decouples_sectors` ($\mathbf{A}_{\mathrm{Lean}}$) shows that diagonal coins decouple all winding sectors: the amplitude in sector s is scaled independently by $e^{i\theta_s}$, with no inter-sector quantum interference.

This is a structural impossibility result: topological localization requires nonlinear coupling between winding sectors, which is provided by the sine-Gordon potential $V(\Phi) = (m_\varphi^2/49)(1-\cos 7\Phi)$ — the defining property of Φ_{MDL}. Linear $\mathbb{Z}_7$-symmetric quantum walks cannot replicate the solitonic structure of Φ_{MDL}, regardless of the choice of coin.

10 Quantum Corrections and Exact S-Matrix

10.1 One-loop effective potential and $\mathbb{Z}_7$ degeneracy

The Coleman–Weinberg effective potential at one loop preserves the $\mathbb{Z}_7$ vacuum degeneracy: $V_{\text{eff}}^{(1)}(\varphi_k) = -2.367 \times 10^{-2}\,\text{GeV}^4$ at all seven $\mathbb{Z}_7$ vacua (identical to within numerical precision; **A**, script `scripts/phimdl_cw_effective_potential.py`). The CW correction to the Higgs quartic $\Delta\lambda_{\text{CW}} = -0.0175$ has the wrong sign to close the Higgs mass gap; the mechanism is ruled out as the λ correction (**A**).

This degeneracy statement is scoped to the pure-φ sector: the loop is computed with $\mathbb{Z}_7$-periodic observables, for which the seven vacua are exactly equivalent. In the full Lagrangian the canonical coupling $V_{\text{coupling}} = \varepsilon|\varphi|^2(D_\mu\chi)^2$ (3) distinguishes the vacuum labels through the χ-sector kinetic normalization $Z_k = 1 + 2\varepsilon\varphi_k^2$ (§2.2); the anomaly-free $\mathbb{Z}_7$ measure theorem and the pure-sector degeneracy stand unchanged.

10.2 Pöschl-Teller fluctuation spectrum

Small fluctuations $\eta = \Phi - \Phi_{\text{kink}}$ about the BPS kink satisfy

$$V_{\text{fl}}(x) = m_\varphi^2\Big[1 - 2\,\text{sech}^2(m_\varphi x)\Big], \tag{35}$$

which is the Pöschl-Teller potential with $s = 1$. Machine-certified: `phimdl_fluctuation_is_poschl_teller` and `arctan_exp_cos_identity` (`PhiMDLFluctuationSpectrum.lean`, zero sorry, $\mathbf{A}_{\text{Lean}}$). The spectrum consists of a single zero mode and the continuum at $k^2 > m_\varphi^2$; the potential is reflectionless. The integrals $\int_{\mathbb{R}} \text{sech}^3 = \pi/2$ and $\int_{\mathbb{R}} \text{sech} = \pi$ are Lean-certified from Mathlib FTC (`integral_sech_cubed`, `integral_sech`, zero sorry, $\mathbf{A}_{\text{Lean}}$).
Quantization in 3+1D: at the Φ_{MDL} coupling $\beta = 7 > \sqrt{8\pi} \approx 5.01$ the 1+1D sine-Gordon quantum theory does not exist (Coleman bound [2]), so the 1+1D semiclassical soliton-mass expansion of Dashen–Hasslacher–Neveu [3] does not apply directly. The physical object is the 3+1D domain wall, whose one-loop tension — and hence the quantum kink mass — is computed in §10.4; the exact 1+1D results enter there as validation benchmarks for the computational machinery.

10.3 Exact S-matrix and TBA kernel

The kink–antikink scattering amplitude is identified with the $\mathbb{Z}\mathbb{Z}$ S-matrix in the sinh-Gordon limit $B \to 1$:

$$S(\theta) = \frac{\sinh\theta - i}{\sinh\theta + i}, \qquad B = 1, \tag{36}$$

machine-certified (**A/D**; script `scripts/phimdl_casimir_tba.py`). The TBA (thermodynamic Bethe Ansatz) kernel follows from the S-matrix:

$$\varphi_{\text{TBA}}(\theta) = \frac{2}{\cosh\theta}, \tag{37}$$

(**A/D**; large-u asymptotics $u \cdot J - \pi/2 \sim -1/u$ verified).

10.4 Quantum kink mass

The one-loop quantum correction to the kink mass is computed from the 3+1D domain-wall tension by channel-resolved dimensional regularization in the interface formalism of Graham, Jaffe, Quandt, and Weigel [4]: each 1D fluctuation channel of frequency Ω contributes the analytic dimensionally regularized transverse energy $-\Omega^3/(12\pi)$, summed over the $\mathbf{A}_{\text{Lean}}$ $s = 1$ Pöschl–Teller spectral data (§10.2) with two Born subtractions and the renormalized two-point-function add-back; the wall tension converts to a kink mass through the area bridge $\Delta M = \Delta\sigma/m_\varphi^2$. The machinery is validated against the exact one-loop sine-Gordon kink mass $-m/\pi$ [3] and the exact φ^4 kink mass $(1/(4\sqrt{3}) - 3/(2\pi))m$, both reproduced to 3×10^{-14}, and by an independent finite-box mode-sum route (sharing no phase-shift machinery) that agrees with the interface result to 3×10^{-5}. The result is

$$\Delta M = \begin{cases} -7.2\,\text{MeV} & \text{(on-shell)}, \\ -10.0\,\text{MeV} & (\overline{\text{MS}} \text{ at } \mu = m_\varphi), \end{cases} \qquad M^Q = 281 \pm 21\,\text{MeV}, \tag{38}$$

(**A**, within a **A/D** analytic framework). The on-shell condition subtracts the meson two-point function at the $m_\varphi = m_\tau$ pole; the central value is the on-shell/$\overline{\text{MS}}$-at-m_φ pair, and the ± 21 MeV band is the envelope over the renormalization-condition family ($\mu \in [m_\varphi/2, 2m_\varphi]$, on-shell, and self-consistent $\mu = 7M^Q$). This band is the irreducible one-loop renormalization-condition freedom of the 3+1D cosine effective field theory: the $(V'')^2$ logarithmic divergence requires a counterterm outside the dimension-≤ 4 potential, so a renormalization condition must be imported; it is fixed only by the $F_{21} \to SU(3)$ ultraviolet completion. The one-loop correction is small ($-3\% \pm 7\%$), and M^Q is consistent with the classical BPS mass $M_{\text{kink}} = 290.10$ MeV within the band. The corrected M^Q keeps the GTE effective-field-theory boundary at $\Lambda_{\text{GTE}} = 7M_{\text{kink}} \approx 2.0$ GeV on both the tree and quantum-mass readings (the seven-kink threshold derivation of Λ_{GTE} is given in P39 [19]; see §10.5).

Two earlier evaluations of M^Q are superseded as incorrect. The $\overline{\text{MS}}$ dimensional-regularization value $M^Q = 321.32 \pm 15.6$ MeV ($\Delta M = +31.22$ MeV) omitted the renormalized diagrammatic add-back, inverted the Levinson phase-shift convention (double-booking $\approx m/12\pi$ of spectral weight), and used a μ-flow ($+m^3/8\pi^2$) corresponding to no consistent scheme (the legitimate flows are $-m^3/6\pi^2$ and $+m^3/12\pi^2$); the finite coefficient formerly reported as a transcendental constant of the $s = 1$ case belongs to this superseded bookkeeping and carries no physical content. The log-UV-cutoff value $M^Q = 230.43$ MeV ($\Delta M = -59.67$ MeV) was a cutoff artifact. Scripts: `scripts/kink_pole_mass_dhn_benchmarks.py`, `scripts/kink_pole_mass_interface_dimreg.py`, `scripts/kink_pole_mass_box_modesum_check.py`, `scripts/kink_pole_mass_branch_verdict.py`.

10.5 Gauge-coupling normalization at the EFT boundary

The gauge coupling at the effective-field-theory boundary is the Sylow–Villain value $e^2(\Lambda_{\text{GTE}}) = 7/2$, an algebraic-certificate (Level-1) rational (`colorCouplingSquared`, $\mathbf{A}_{\text{Lean}}$). Its comparison with experiment is a Level-2 continuum statement and is scheme-aware throughout: the GTE value is defined in the Villain (substrate) convention, while the PDG running coupling is $\overline{\text{MS}}$.

The boundary scale. The EFT boundary is derived, not calibrated: $\Lambda_{\text{GTE}} = 7\,M_{\text{kink}}^{\text{phys}}$, where the multiplier $7 = |\mathbb{Z}_7|$ counts the minimal $\mathbb{Z}_7$-neutral kink chain — the full-period unwinding threshold of the non-compact Φ (exact in the repulsive channel; the derivation is given in P39 [19]). The tree reading gives $\Lambda_{\text{GTE}} = 7\,(8/49)m_\tau = (8/7)m_\tau = 2030.70 \pm 0.14$ MeV; the quantum-mass reading gives $7M^Q = 1.970 \pm 0.146$ GeV (eq. (38)). The two readings agree within 0.5σ, so the boundary is the single scale $\Lambda_{\text{GTE}} \approx 2.0$ GeV with envelope 1.96 ± 0.15 GeV.

The comparison. Running PDG $\alpha_s(M_Z) = 0.1180 \pm 0.0009$ (PDG 2024) down to the boundary gives $e^2_{\overline{\rm MS}}(\Lambda_{\rm GTE}) = 3.74 \pm 0.08$ (tree reading) / 3.79 ± 0.16 (quantum-mass reading), versus the GTE value $7/2 = 3.5$: a +7–8% offset, i.e. $+3.0\sigma$ / $+2.0\sigma$ before any matching constant. This offset is *not* attributable to a known scheme conversion: the derived conversion coefficient for the literal heat-kernel (Villain) lattice action is 60.8 — nonperturbative at $g^2 = 7/2$ — while gradient-flow and proper-time completions have the wrong sign. The honest decomposition of the one-loop matching constant is: (i) the gauge sector contributes zero exactly (the abelian Villain spin-wave sector is Gaussian); (ii) the coset-sector threshold constant, derived by standard effective-gauge-theory matching [5, 26], is $c_{\rm coset} \in [-2.51, -1.00]$ for all EFT-coherent coset-gap readings — the *wrong sign* (it deepens the offset); (iii) the kink vacuum polarization $c_{\rm kink}$ is sign-positive on every gauge-invariant ultraviolet completion (derived; the kink is a composite of neutral constituents, and its charged loop loses ultraviolet support at the dissolution scale), with a lattice-tape control value of -1.00 showing that tape discreteness alone cannot carry the constant.

The measured dissolution constant. The remaining unknown — the kink dissolution scale — has been measured non-perturbatively: a substrate-regulated lattice Monte Carlo of the 3+1D domain wall at the physical tape spacing $a = 1/\Lambda_{\rm GTE}$ (bare cosine coefficient tuned to the meson pole mass measured in the selected vacuum; one-kink-sector projection; estimator chain validated on the exact free-field dispersion, on the classical lattice kink with the identical estimator, and on a 1+1D dissolution positive control; transverse boxes $L = 10$–28 with a capillary-geometry infinite-volume completion) gives a charge-distribution broadening

$$b \;\equiv\; \frac{r^{\rm meas}_{\rm RMS}}{r_{\rm class}} \;=\; 1.189 \pm 0.015\,({\rm stat}) \pm 0.047\,({\rm syst}) \tag{39}$$

relative to the classical sech kink (**A**) — a dissolution scale $\Lambda_{\rm diss} = m_\varphi/b = 1495 \pm 61\,{\rm MeV} = 0.84\,m_\varphi$. The corresponding matching constant $c_{\rm kink} = 8\ln(\Lambda_{\rm GTE}/\Lambda_{\rm diss}) = +2.45 \pm 0.33$ (tree) / $+2.21 \pm 0.33$ (quantum mass) reduces the $e^2(\Lambda_{\rm GTE}) = 7/2 \leftrightarrow$ PDG offset from $+3.0\sigma/+2.0\sigma$ to

$$+1.5\sigma\ \text{(tree)} \qquad / \qquad +1.3\sigma\ \text{(quantum mass)}: \tag{40}$$

the genuine quantum broadening of the wall closes roughly half the one-loop matching offset, and the residual is consistent with the uncomputed two-loop matching term. The quantum wall profile remains a sech to fit residual 0.004, and the quantum wall tension is ≈ 0.42 of the classical value (meson-cloud softening).

The tape spacing. The physical tape spacing $a = 1/\Lambda_{\rm GTE}$ used by the lattice campaign above is itself derived, not imposed: Minimum Description Length saturation of the substrate tape selects exactly this spacing — the Tape Saturation Theorem of P50 [23] (§Continuum Limit and the Physical Point), a bridge statement calibrating the Level-1 certificate against the Level-2 continuum spectrum (**A/D**, conditional on one named premise, the Compton-Support Criterion; machine-certified at that conditional grade as `tape_saturation_theorem` in the P50 Lean inventory). The named criterion is irreducible from the algebraic certificate alone and provably minimal: any derivation of the saturation condition must supply content logically equivalent to it (`hosting_boundary_csc_equivalence`, **A/D**). At the saturated spacing the lattice dictionary interlocks with zero tuning — $aM_{\rm kink} = 1/7$ and $am_\varphi = 7/8 = (49/8)\cdot(1/7)$, exactly the bare meson mass employed by the campaign, now derived rather than chosen — and the correlation length of the substrate's thermal shadow (the spin-7 chain of P50) at the physical point equals $\xi^* = |\mathbb{Z}_7| = 7$ lattice sites exactly. This yields a falsifiable substrate target: a Monte Carlo measurement of the kink–kink correlation length at the physical point must find $\xi_{\mathrm{kink}} = 7$ cells (mixed-reading band 6.78–7.23 $\pm$ 0.54).

Structural nulls and verdict. Two pre-registered structural checks pass: 7/2 is uniquely closest to the PDG-run value within the Sylow rational family $\{7/1, 7/2, 7/3, 7/6, 3/2, 5/2, 9/2, 21/2\}$, with margin $\times 2.9$ to the runner-up (`sylow_family_pdg_match_unique_closest`, $\mathbf{A}_{\mathrm{Lean}}$); and the exact-match scale $\mu^* = 2.357$ GeV coincides with Λ_{GTE} rather than M_Z (matching at M_Z is off by $\times 2.4$), consistent with the $F_{21} \to SU(3)$ Burnside completion ending the EFT at Λ_{GTE}. The overall verdict is a **A/D**-consistency: the $e^2(\Lambda_{\mathrm{GTE}}) = 7/2$ identification is consistent with PDG running at the 1.3–1.5σ level with every scale derived and the dominant matching constant measured; no anomaly is established, and the residual sits at the expected size of the two-loop matching term. The derived bracket is $e^2(\Lambda_{\mathrm{GTE}}) \in [3.50, 3.76]$, whose two ends are independent derivation chains; a future tightening of the measured broadening together with a computed two-loop matching term makes the comparison decisive in either direction. Claim levels: formulation and sign positivity of c_{kink}, **A/D**; the c_{kink} value and the broadening measurement, **A**. Scripts: `scripts/kink_form_factor_precision_runner.py`, `scripts/kink_form_factor_precision_extraction.py`, `scripts/kink_form_factor_precision_coupling_verdict.py`, `scripts/kink_vacuum_polarization_matching_constant.py`, `scripts/burnside_threshold_coset_matching.py`, `scripts/color_coupling_e_normalization.py`.

10.6 Kink-sector contribution to the hadronic vacuum polarization

The measured kink charge form factor yields a parameter-free prediction for the kink-sector contribution to the running of the electromagnetic coupling. The kink carries the Cartan charge content $t_{\mathrm{kink}} = 3$ (weights $\{0, \pm 1/2\}$ over six Dirac flavors; a Level-1 certificate, `kink_cartan_charge_trepresentation`, $\mathbf{A}_{\mathrm{Lean}}$), with charge density $\rho_{\mathrm{kink}}(r) = (b/2)\, m_\varphi^2 \,\mathrm{sech}^2(m_\varphi r/b)$ — the sech profile of Theorem 5.3 with the measured broadening (39). Inserting the corresponding spectral density into the standard dispersive integral for the hadronic vacuum polarization [1], $\Delta\alpha_{\mathrm{kink}} = (\alpha/3\pi) \int_{4M^2_{\mathrm{kink}}}^{\infty} \mathrm{Im}\, F_{\mathrm{kink}}(s)\, s^{-1}\, ds$, gives

$$\Delta\alpha_{\mathrm{kink}} = \begin{cases} (3.47 \pm 0.23) \times 10^{-4} & \text{(single-pole spectral model; error from } b), \\ 3.70 \times 10^{-4} & \text{(extended spectral model)}, \end{cases} \tag{41}$$

(**A**) at the $\Delta\alpha_{\mathrm{had}}$ reference scale — 1.26–1.34% of the total hadronic vacuum-polarization contribution, and 4.67–4.98% of the non-perturbative 1–2 GeV window that dominates the residual between the GTE hadronic vacuum polarization and the experimental $(g-2)_\mu$ central value (the muon $g-2$ analysis is given in P39 [19]). These numbers are first-principles predictions derived without external $R(s)$ data: the spectral weight sits at the measured dissolution scale $\Lambda_{\mathrm{diss}} = 1.495$ GeV, inside the non-perturbative window. The dominant $\rho/\omega/\phi$ contribution to that window is not derived here and remains an open problem. The prediction is falsifiable against future lattice and dispersive determinations of the hadronic vacuum polarization as the kink mass and charge radius become better constrained.

The same kink spectral function, integrated against the standard $(g-2)_\mu$ HVP kernel $\hat{K}(s) = \int_0^1 du\, u^2(1-u)/[u^2 + (s/m_\mu^2)(1-u)]$ [1], gives a first-principles contribution to the muon anomalous magnetic moment from the Φ_{MDL} kink sector:

$$a_\mu^{\mathrm{kink}} = \begin{cases} (46.6 \pm 17.1) \times 10^{-11} & \text{(scalar loop model)}, \\ 65.9 \times 10^{-11} & \text{(vector model)}, \end{cases} \tag{42}$$

(**D**), covering 18.7%–26.4% of the 4σ anomaly gap. This is a parameter-free first-principles partial-window contribution. The Φ_{MDL} kink-sector contribution is a BSM term, complementary to the standard QCD hadronic vacuum polarization: the kink spectral function is smooth and continuous ($\rho_{\mathrm{kink}} \propto \beta^3$ above the pair threshold $\sqrt{s_{\mathrm{pair}}} = 2M_{\mathrm{kink}} \simeq 578$ MeV) with no resonance poles, distinguishing it spectroscopically from the sharp Breit–Wigner resonances of the standard hadronic R-ratio. Accordingly a_μ^{kink} is an additional BSM-sector contribution to $(g-2)_\mu$; it does not replace or recompute the dominant $\rho(770)/\omega(782)/\phi(1020)$ hadronic contribution, which is not derived in the present framework and remains an open problem (OQ-HVP-1).

Script: `scripts/kink_form_factor_delta_alpha_had.py`.

11 Conclusion

We have established the quantum field theory of Φ_{MDL} as the continuum companion to the Three-Layer Chiral Minkowski CA (P41). The principal results are:

1. **Poincaré invariance** ($\mathbf{A}_{\mathrm{Lean}}$): the Φ_{MDL} KG dispersion is Lorentz-invariant (machine-certified in `LorentzInvariance.lean`); the stress-energy tensor transforms as a rank-2 tensor (**A/D**, 400 random boosts verified).
2. **Born rule from field amplitudes** ($\mathbf{A}/\mathbf{D}/\mathbf{A}_{\mathrm{Lean}}$): $P(x) = |\partial_x\Phi|^2/\int|\partial_x\Phi|^2$ normalizes exactly to 1; extended to overlapping kinks via the dual-frame inner product (`DualFrameBornRule.lean`); [**D**]-collapse implements projective measurement onto the kink eigenbasis.
3. **Physical quantum state** ($\mathbf{A}_{\mathrm{Lean}}$ conditional): the pre-measurement state is the Hamiltonian thermal ensemble on PSC-admissible sectors $\{0,2,3,4,6\}$ (`PhiMDLThermalState.lean`); vacuum-dominant at observed cosmic temperatures; uniform at high T.
4. **3+1D extension** ($\mathbf{A}/\mathbf{D}/\mathbf{A}_{\mathrm{Lean}}$): kinks become planar domain walls with tension $\sigma = 290.10\,\mathrm{MeV\,GeV^{-2}}$; $\mathbb{Z}_7$ superselection and Born rule are unchanged; SR time dilation is exact. When two walls intersect, they form a junction line with exact tension $\lambda_{\mathrm{dim}} = -16/49$ ($\mathbf{A}_{\mathrm{Lean}}$; $\lambda = -1654.77\,\mathrm{MeV\,fm}$; attractive, $|\lambda/\sigma| = 2/m_\varphi$).
5. **Explicit descent map** (**A**): the Cook A-glider $\mathbb{Z}_7$ winding profile matches the BPS kink with RMSD $= 5.34\% < \varepsilon_0(7) = 6.71\%$ and Pearson $r = 0.994$, providing direct numerical confirmation of the CMCA$\to$ Φ_{MDL} continuum limit.
6. **No-finite-CA uniqueness** ($\mathbf{A}_{\mathrm{Lean}}$): every finite-$M$ discrete model has strictly positive Lorentz error; Φ_{MDL} is the unique zero-error model in the $\mathbb{Z}_7$-symmetric class (`phimdl_is_unique_exact_lorentz_model`).
7. **QCA impossibility** ($\mathbf{A}_{\mathrm{Lean}}$): any $\mathbb{Z}_7$-winding-conserving coin must be diagonal, decoupling all sectors and precluding quantum interference; topological localization is structurally impossible in a linear quantum walk (`commutes_with_winding_iff_diagonal`).
8. **Algebraic Lifting and Descent** ($\mathbf{A}_{\mathrm{Lean}}$): the SM winding sector lifts to [**D**]-weighted states on Φ_{MDL} (`algebraic_lifting_theorem`); all F_{21} conservation laws descend to any continuum realisation at every $M \geq 1$ (`algebraic_descent_theorem`). The SR Nyquist correction $\varepsilon_0(M) = \pi^2/(3M^2)$ is M-dependent and does not descend (`sr_accuracy_depends_on_M`).
9. **Continuum limit** ($\mathbf{A}_{\mathrm{Lean}}$ conditional): $\varepsilon_0(M) = \pi^2/(3M^2) \to 0$ as $M \to \infty$; Φ_{MDL} is the unique terminal Level-1 GTE object (`CMCAContinuumLimit.lean`).
10. **Quantum corrections** (§10): Pöschl-Teller spectrum $s = 1$ ($\mathbf{A}_{\mathrm{Lean}}$; `phimdl_fluctuation_is_poschl_teller`); $\mathbb{Z}_7$ one-loop degeneracy preserved in the pure-φ sector (**A**); ZZ S-matrix $S(\theta) = (\sinh\theta - i)/(\sinh\theta + i)$ (**A/D**); TBA kernel $\varphi(\theta) = 2/\cosh\theta$ (**A/D**); quantum kink

mass $M^Q = 281 \pm 21$ MeV (**A**; interface dimensional regularization, benchmark-validated, two independent routes; $\Delta M = -7 \ldots -10$ MeV, consistent with the classical BPS mass within the scheme band).

11. **Coupling-sector completion** (§2.2, §10.5): the full Lagrangian carries the canonical coupling $V_{\text{coupling}} = \varepsilon|\varphi|^2(D_\mu\chi)^2$ with $\varepsilon = 7/2$, the χ-sector mass parameter $g = m_\varphi = m_\tau$ (**B**), and the gauge coupling $e^2(\Lambda_{\text{GTE}}) = 7/2$ ($\mathbf{A}_{\text{Lean}}$ algebraic value); the $\mathbb{Z}_7$ vacuum degeneracy is exact in the pure-φ sector while V_{coupling} distinguishes the vacua through the χ-kinetic normalization. With the non-perturbatively measured kink dissolution constant ($b = 1.189 \pm 0.049$, **A**), the $e^2 = 7/2$ identification is consistent with PDG running at $+1.5\sigma$ (tree) / $+1.3\sigma$ (quantum kink mass) — no anomaly.
12. **Kink-sector hadronic vacuum polarization** (§10.6): the measured kink charge form factor yields a parameter-free prediction $\Delta\alpha_{\text{kink}} = (3.5\text{–}3.7) \times 10^{-4}$, i.e. 1.3% of the total hadronic vacuum-polarization contribution (**A**). The same kink spectral function gives a first-principles BSM contribution to $(g-2)_\mu$: $a_\mu^{\text{kink}} = (46.6 \pm 17.1) \times 10^{-11}$ (scalar) / 65.9×10^{-11} (vector) (**D**), covering 18.7–26.4% of the 4σ anomaly gap and spectroscopically distinct from the standard $\rho/\omega/\phi$ sector (no resonance poles in the kink spectral function).

The two-function picture—Born rule $\mathcal{B}$ and transputation selector $\mathbb{P}^\top$—provides a complete, axiom-minimal account of quantum measurement in the GTE framework, machine-certified in Lean 4 with zero sorry (conditional on the named [**D**] axioms of P34).

A Lean 4 Theorem Inventory

All theorems listed below are in the `ugp-lean` repository (canonical public repo). Zero sorry. Disclosed axioms follow the theorem list.

Table 3: Lean 4 theorem inventory for P42. †: conditional on named axioms; see text.

Theorem name	Module	S	Content
`phimdl_kink_masses_equal`	`Universality/PhiMDLThermalState`	0	All $\mathbb{Z}_7$ kink masses equal
`psc_admissible_sectors`	`Universality/PhiMDLThermalState`	0	$\{0, 2, 3, 4, 6\}$ admissible
`psc_admissible_card`	`Universality/PhiMDLThermalState`	0	Card $= 5$
`psc_sectors_partition`	`Universality/PhiMDLThermalState`	0	Admissible $\cup$ forbidden $= \mathbb{Z}_7$
`thermal_state_sector_prob`	`Universality/PhiMDLThermalState`	0	$P(k \mid T) = e^{-M_k/T}/Z_T$
`sectorProb_admissible_sum`	`Universality/PhiMDLThermalState`	0	Normalization
`vacuum_particle_prob_ratio`	`Universality/PhiMDLThermalState`	0	Vacuum dominance $T \ll M_{\text{kink}}$
`phimdl_thermal_state_master`	`Universality/PhiMDLThermalState`	0	Master bundle thermal state
`dual_gram_identity`	`Universality/DualFrameBornRule`	0	$GG^{-1} = \mathbf{1}$
`dual_frame_recovers_coefficients`	`Universality/DualFrameBornRule`	0	$a_k = c_k$
`born_rule_from_dual_frame`	`Universality/DualFrameBornRule`	0	$\sum_k \lvert a_k\rvert^2 = 1$

(continued on next page)

(Table 3 continued)

Theorem name	Module	S	Content
born_rule_from_dual_frame_master	Universality/DualFrameBornRule	0	Dual-frame Born master bundle
transputation_is_minimizer	Substrate/TransputationStateSelector	0	$\mathbb{P}^{\top} = \arg\min D(\rho \mid w)$
transputation_minimizer_unique	Substrate/TransputationStateSelector	0	Uniqueness of minimum
two_function_picture	Substrate/TransputationStateSelector	0	$\mathcal{B} \circ \mathbb{P}^{\top}_{D}$
transputation_state_selector_bundle	Substrate/TransputationStateSelector	$0^{\dagger}$	Two-function selector bundle
poincare_invariance_of_kg	Universality/LorentzInvariance	0	Full Poincaré
phimdl_lorentz_correction_tendsto_zero	Framework/CMCAContinuumLimit	0	$\varepsilon_0(M) \to 0$ as $M \to \infty$
cmca_continuum_limit_is_phimdl	Framework/CMCAContinuumLimit	$0^{\dagger}$	CMCA $\to \Phi_{\mathrm{MDL}}$ master
born_rule_unconditional	Universality/BeableWindingPartitionInstance	0	Sector Born rule
algebraic_lifting_theorem	Spacetime/LiftingTheorem	0	PSC-admissible configs realised in f_{MDL} (P28)
algebraic_descent_theorem	Universality/AlgebraicDescentTheorem	0	M-independence of F_{21} content (P28)
sr_accuracy_depends_on_M	Universality/AlgebraicDescentTheorem	0	$\varepsilon_0(M)$ is M-dependent (P28/P41)
mkink_from_scc	Universality/GUTStructure	0	$M_{\mathrm{kink}} = (8/49)m_{\tau}$ (P34/P35)
phimdl_domain_wall_junction_tension_exact	Substrate/WindingCoinDecoupling	0	$\lambda_{\mathrm{dim}} = -16/49$ (exact)
phimdl_junction_is_attractive	Substrate/WindingCoinDecoupling	0	$\lambda < 0$: walls attract at junction
phimdl_junction_to_wall_ratio	Substrate/WindingCoinDecoupling	0	$\lvert\lambda/\sigma\rvert = 2/m_{\varphi}$
commutes_with_winding_iff_diagonal	Substrate/WindingCoinDecoupling	0	$\mathbb{Z}_7$-winding-commuting coin iff diagonal
diagonal_coin_decouples_sectors	Substrate/WindingCoinDecoupling	0	Diagonal coin decouples all $\mathbb{Z}_7$ sectors
no_finite_ca_exact_lorentz_replica	Framework/CMCAContinuumLimit	0	$\varepsilon_0(M) > 0$ for all finite M
phimdl_is_unique_exact_lorentz_model	Framework/CMCAContinuumLimit	0	Φ_{MDL} unique zero-error Lorentz model
phimdl_axial_current_topological	Substrate/ChiralCurrentL2	0	$\partial_{\mu} J^{\mu}_{A} = 0$ (Schwarz)
phimdl_vector_current_topological	Substrate/ChiralCurrentL2	0	$\partial_{\mu} J^{\mu}_{V} = 0$ (Schwarz)
va_fraction_lifts_to_l2_chiral_current	Substrate/ChiralCurrentL2	0	$32/125$ lifts to L2 J^{μ}_{A}

(continued on next page)

(Table 3 continued)

Theorem name	Module	S	Content
phimdl_l2_chiral_current_bundle	Substrate/ChiralCurrentL2	0	G16 chiral-current bundle
phimdl_weak_charged_current	Algebra/GaugeMDL	0	J_W^μ from $\lvert D_\mu \Psi \rvert^2$ (MDL gauging)
weak_charged_current_bilinear_antisymmetric	Algebra/GaugeMDL	0	Isospin-exchange antisymmetry of J_W^μ bilinear
phimdl_l2_chiral_and_weak_current_bundle	Substrate/ChiralCurrentL2	0	G16+G18: V–A + weak charged current
vcoupling_breaks_z7_shift	Physics/ZSevenVacuumSelection	0	V_{coupling} not invariant under $\mathbb{Z}_7$ vacuum shift
z7_vacua_chi_kinetic_inequivalent	Physics/ZSevenVacuumSelection	0	$Z_k = 1 + 2\varepsilon\varphi_k^2$ distinguishes vacua
wall_bias_minimum_unique	Physics/ZSevenVacuumSelection	0	Vacuum-label bias has unique minimum
vcoup_periodic_profile_fails_bps_window	Physics/ZSevenVacuumSelection	0	Coupling-form discriminator: periodic profile fails BPS window
z7_coupling_seam_discontinuity	Physics/ZSevenVacuumSelection	0	$\mathbb{Z}_7$-shift-invariant completion has seam discontinuity
compact_completion_minimum_at_k0	Physics/ZSevenVacuumSelection	0	Compact completion minimized at $k^* = 0$
scalar_cw_step_inward	Physics/ZSevenVacuumSelection	0	Scalar one-loop step direction inward
boundary_sensitivity_coefficient_positive	Physics/ZSevenVacuumSelection	0	Boundary-matching sensitivity coefficient > 0
sylow_family_pdg_match_unique_closest	Universality/SylowIndexCouplingHierarchy	0	$7/2$ uniquely closest in the Sylow rational family
thermal_leading_inward_dominance	Universality/SylowIndexCouplingHierarchy	0	Thermal leading term dominates inward
kink_vacuum_polarization_algebraic_core	Physics/KinkVacuumPolarization	0	Algebraic core of the VP matching constant
kink_cartan_charge_trepresentation	Physics/KinkVacuumPolarization	0	Kink Cartan charges $\{0, \pm 1/2\}$; $t_{\text{kink}} = 3$
b_hat_continuity_identity	Physics/KinkVacuumPolarization	0	$\hat{b}_{\text{kink}} = -4$ continuity with $b_0 = 7$
c_kink_s1tree_rational_log	Physics/KinkVacuumPolarization	0	$c_{\text{kink}}(\text{tree}, \text{PV}) = 8\ln(8/7)$
kink_positivity_inequality	Physics/KinkVacuumPolarization	0	Sign positivity: $\gamma_E > 2\ln(7/8)$
sech_form_factor_moment_bundle	Physics/KinkFormFactor	0	sech^2 form-factor moment identities
sech_autocorrelation_identity	Physics/KinkFormFactor	0	Potential autocorrelation closed form

Lean build verification. All modules compile with:

```
cd ugp-lean
```

```
lake build UgpLean.Universality.PhiMDLThermalState
lake build UgpLean.Universality.DualFrameBornRule
lake build UgpLean.Substrate.TransputationStateSelector
lake build UgpLean.Framework.CMCAContinuumLimit
lake build UgpLean.Universality.LorentzInvariance
lake build UgpLean.Universality.BeableWindingPartitionInstance
lake build UgpLean.Spacetime.LiftingTheorem
lake build UgpLean.Universality.AlgebraicDescentTheorem
lake build UgpLean.Substrate.WindingCoinDecoupling
lake build UgpLean.Substrate.ChiralCurrentL2
lake build UgpLean.Algebra.GaugeMDL
lake build UgpLean.Physics.ZSevenVacuumSelection
lake build UgpLean.Physics.KinkVacuumPolarization
lake build UgpLean.Physics.KinkFormFactor
lake build UgpLean.Universality.SylowIndexCouplingHierarchy
```

Sorry count: zero (for all theorems in Table 3). Disclosed axioms: `d2_universal` (D2 of P34, PSC Presentation Invariance in universal form) in `PSCPILorentzMain.lean` and `CMCAContinuumLimit.lean`; `architectural_restriction` in `CoherenceMeasureUniqueness.lean`; the named physical-interpretation axioms of `GTECategoryStructure.lean`. The Born-rule modules `BeableWindingPartitionInstance.lean`, `DualFrameBornRule.lean`, and `PhiMDLThermalState.lean` carry zero custom physics axioms. The `WindingCoinDecoupling.lean` module contains two auxiliary `sorry`-carrying *lemmas* (`bps_kink_integral_eq_four` and `symmetry_integral_vanishes`) that are used only for internal documentation; all theorems cited in Table 3 are proved without these lemmas and carry zero sorry. The `ZSevenVacuumSelection`, `KinkVacuumPolarization`, and `KinkFormFactor` rows certify the algebraic cores of the coupling-sector results of §2.2 and §10.5; the corresponding continuum-field conclusions carry the **A/D/A** labels stated there. A formal certification of the dissolution Born-floor bound remains open.

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